

DRAFT BASIC ASSESSMENT REPORT

FOR THE

ADDITION OF BM LINE 3 TO THE EXISTING AUTO-X OPERATIONS, NEAVE TOWNSHIP (KORSTEN) IN GQEBERHA WITHIN THE NELSON MANDELA BAY MUNICIPALITY, EASTERN CAPE PROVINCE

DEDEAT Reference No: ECm1/C/LN1/M/36-2026

24 August 2026

Prepared By:



**ABANTU
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BASIC ASSESSMENT REPORT

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Basic Assessment Report in terms of the Environmental Impact Assessment Regulations, 2014 as amended, promulgated in terms of the National Environmental Management Act 107 of 1998, as amended.

Kindly note that:

1. This **basic assessment report** (BAR) is a standard report that may be required by a competent authority (CA) in terms of the Environmental Impact Assessment Regulations, 2014 as amended (the Regulations) and is meant to streamline applications. Please make sure that it is the report used by the particular CA for the activity that is being applied for. This report is current as of **4 May 2026**. It is the responsibility of the applicant to ascertain whether subsequent versions of the form have been published or produced by the CA.
2. The report must be typed within the spaces provided in the form. The size of the spaces provided is not necessarily indicative of the amount of information to be provided. The report is in the form of a table that can extend itself as each space is filled with typing.
3. Where applicable **tick** the boxes that are applicable or **black out** the boxes that are not applicable in the report.
4. An incomplete report may be returned to the applicant for revision.
5. The use of "not applicable" in the report must be done with circumspection because if it is used in respect of material information that is required by the CA for assessing the application, it may result in the rejection of the application as provided for in the Regulations.
6. This report must be handed in at offices of the relevant CA as determined by each authority **unless indicated otherwise by the Department**.
7. No faxed or e-mailed reports will be accepted **unless indicated otherwise by the Department**.
8. The report must be compiled by an independent environmental assessment practitioner (EAP). The EAP must satisfy condition 11 below.
9. Unless protected by law, all information in the report will become public information on receipt by the CA. Any interested and affected party (I&AP) should be provided with the information contained in this report on request, during any stage of the application process.
10. A CA may require that for specified types of activities in defined situations only parts of this report need to be completed.
11. EAP Conditions:
 - 11.1 The EAP must be registered in terms of S24H Regulations with the Registration Authority the Environmental Assessment Practitioners Association of South Africa (EAPASA) as from 8 August 2022.
 - 11.2 S24H (14) states that "only a person registered as an EAP may perform tasks in connection with an application for an environmental authorisation (EA) contemplated in: (a) Chapter 5 of the National Environmental Management Act 107 of 1998, as amended (the Act) read with the Regulations; (b) Section 24G of the Act; and (c) Chapter 5 of the National Environmental Management Waste Act 2008 (Act No 59 of 2008) as amended (NEMAQA) read with the Regulations.
 - 11.3 Tasks in regulation 14 may only be conducted by an EAP that is registered.
 - 11.4 Regulations 20 of S24H indicates the offences and penalties as indicated below: "20. Offences and penalties: (1) A person is guilty of an offence if that person: (a) contravenes regulation 14 of the Regulations; or (b) pretends to be a registered environmental assessment practitioner (Reg. EAP) or registered candidate environmental assessment practitioner. (2) A person convicted of an offence in terms of subregulation (1) is liable to the penalties contemplated in section 49B(3) of the Act." Section 49B(3) of the Act states: "A person convicted of an offence in terms of section 49A(1)(h), (l), (m), (n), (o) or (p) is liable to a fine or to imprisonment for a period not exceeding one year, or to both a fine and such imprisonment."

UNDERTAKING (AFFIRMATION BY THE EAP)

Abantu Environmental Services (Pty) Ltd (Abantu) hereby confirms that, to the best of our knowledge, the information provided in this report was correct at the time of compilation. Information included in this report was based on the information which was provided to us by the Applicant and the commissioned specialist. It is further confirmed that all comments received from stakeholders and I&APs will be included in the final BAR to be submitted to the CA for review and decision-making.

ASSUMPTIONS AND GAPS IN KNOWLEDGE

The following assumptions and potential gaps in knowledge apply:

- All information provided by the applicant to the EAP was correct and valid at the time it was provided.
- The EAP does not accept any responsibility in the event that additional information comes to light at a later stage after the process.
- All data from unpublished research is valid and accurate at the time this report was written.
- The scope of this investigation is limited to assessing the potential environmental impacts associated with the proposed addition of a third battery machine line (BM Line 3) for the assembly of lead-acid batteries.

It should be noted that findings, recommendations, and conclusions provided in this report are based on the author's best scientific and professional knowledge, and experience. No part of this report may be amended or extended without prior written consent of the author. Any recommendations, statements or conclusions drawn from or based on this report must clearly cite or refer to this report. Whenever such recommendations, statements, or conclusions form part of the main report to current investigation, this report must be included in its entirety.

ACKNOWLEDGEMENTS

The authors acknowledge Auto-X for their assistance with project information, facility illustration, and the associated project background information documents as well as for responding to technical queries related to the project.

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Acronyms

Abantu	Abantu Environmental Services (Pty) Ltd
AEL	Atmospheric Emission Licence
Auto-X	Auto X – P.E. Manufacturing Facility, trading as Auto-X (Pty) Ltd
BA	Basic Assessment
BAR	Basic Assessment Report
BM Line 3	Battery Machine Line 3
BTech	Baccalaureus Technologiae
CA	Competent Authority
CBA	Critical Biodiversity Area
CESA	Critical Ecological Support Area
cm	Centimetre
CRR	Comments and Responses Report
DEDEAT/ the Department	Eastern Cape Department of Economic Development, Environmental Affairs and Tourism
EA	Environmental Authorisation
EAP	Environmental Assessment Practitioner
EAPASA	Environmental Assessment Practitioners Association of South Africa
ECBCP	Eastern Cape Biodiversity Conservation Plan
EMPr	Environmental Management Programme
EMS	Environmental Management System
ESA	Ecological Support Area
GDP	Gross Domestic Product
GHG	Greenhouse Gas
GN	General Notice
HEPA	High-Efficiency Particulate Air (filter)
H ₂ SO ₄	Sulfuric acid
IDP	Integrated Development Plan
IHS	International Hearing Society
ISO	International Standardisation Organisation
I&AP	Interested and Affected Party
kg/a	Kilogram per annum
km	Kilometre
LAQS	Lethabo Air Quality Specialists (Pty) Ltd
LPG	Liquified Petroleum Gas
m	Metre
m ²	Square metre
m ³	Cubic metre
mm	Millimetre
mg/Nm ³	Milligram per cubic nanometre
NAAQS	National Ambient Air Quality Standards published under Government Notice 1210 of 2009
NEMA	National Environmental Management Act 107 of 1998 as amended
NEMAQA	National Environmental Management Air Quality Act 39 of 2004 as amended
NEMWA	National Environmental Management Waste Act 59 Of 2008 as amended

NFEPA	National Freshwater Ecosystem Priority Area
NMBM	Nelson Mandela Bay Municipality
NO ₂	Nitrogen dioxide
NO _x	Nitrogen oxides
PA	Protected Area
Pb	Lead
PbO	Lead oxide
PM	Particulate matter
PPE	Personal protective equipment
PPP	Public participation process
Pr.Sci.Nat	Professional Natural Scientist
Reg. EAP	Registered Environmental Assessment Practitioner
REL	Reference Exposure Levels
RSA	Republic of South Africa
SACNASP	South African Council for Natural Scientific Professions
SAHRA	South African Heritage Resources Agency
SANBI	South African National Biodiversity Institute
SANS	South African National Standards
SO ₂	Sulphur dioxide
SO ₃	Sulphur trioxide
SSV	Site Sensitivity Verification
StatsSA	Statistics South Africa
S&EIA	Scoping and Environmental Impact Assessment
The Act	National Environmental Management Act 107 of 1998 as amended
The Regulations	Environmental Impact Assessment Regulations, 2014 as amended
TVOCs	Total Volatile Organic Compounds
WUL	Water Use Licence
°C	Degrees Celsius
µg/m ³	Micrograms per cubic metre

SECTION A: ACTIVITY INFORMATION

Has a specialist been consulted to assist with the completion of this section?

YES

If YES, please complete form XX for each specialist thus appointed:

Any specialist reports must be contained in **Appendix D**.

1. ACTIVITY DESCRIPTION

Describe the activity, which is being applied for, in detail

1.1. Introduction and Proposal

Auto X – P.E. Manufacturing Facility, trading as Auto-X (Pty) Ltd (hereinafter referred to as Auto-X) is a holder of an Atmospheric Emission Licence (AEL), reference number 19/2/9/2/1/2/ NMBM AEL 13/36(02), in terms of section 45(1) of the NEMAQA for the production (manufacturing and assembly) of lead-acid batteries. The licence was issued for lead (Pb) smelting (sub-category 4.13) and the production of acids (sub-category 7.2) as part of battery manufacturing (creation of individual components) and assembly (putting of the components together) within the Auto-X facility. The facility is situated in Korsten, Gqeberha under the Nelson Mandela Bay Municipality (NMBM) as presented in **Figure 1**.

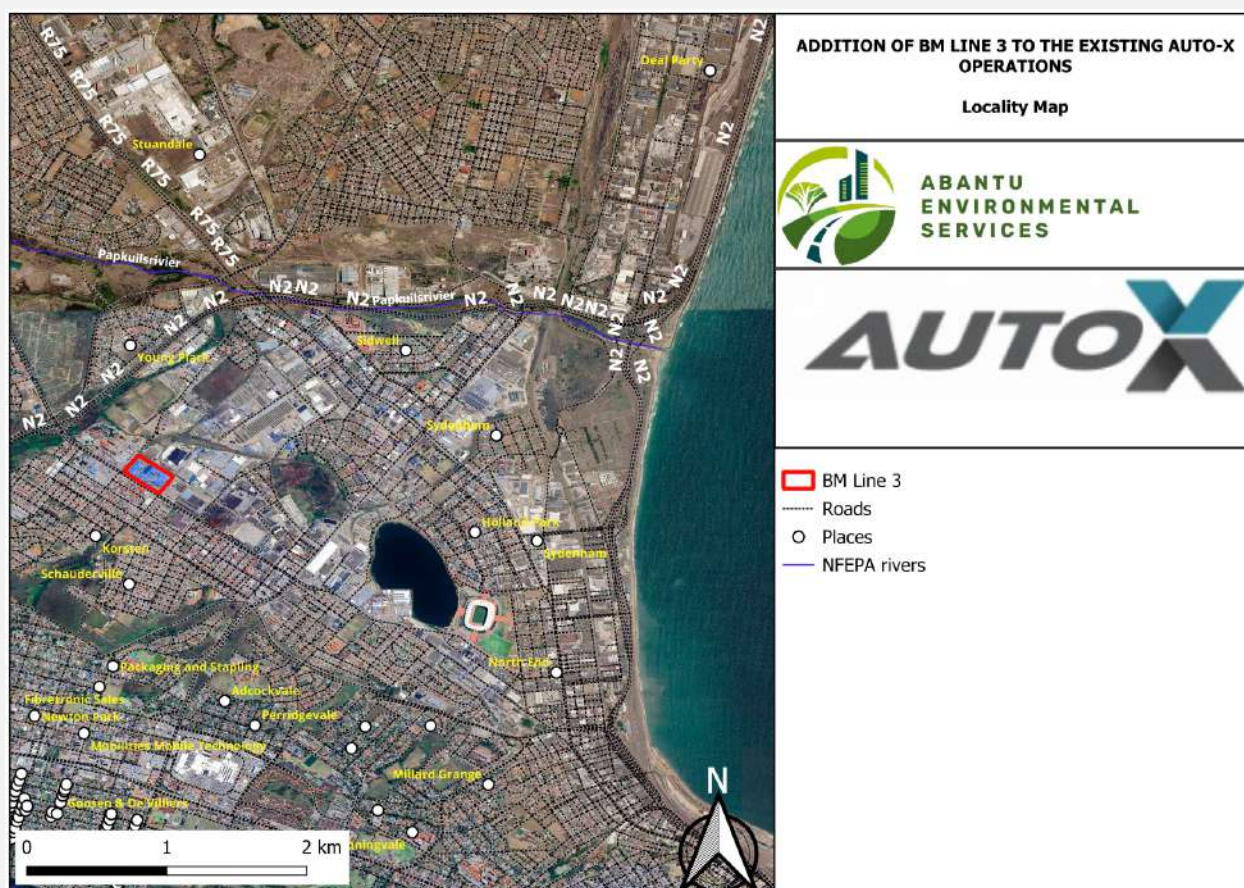


Figure 1: Locality of the existing operation and proposed addition

The applicant is proposing to install and operate BM Line 3, along with stack AX4, at its Korsten facility. This line is an identical replica of the existing, approved BM Line 1 or 2 and will be served by a new Donaldson bag filter, identical to those already in use. This is envisaged to align with the retirement of an existing line, and planned use of latest machinery/ technology in battery assembly. Thus, alterations are to be made but installation of equipment.

Given off by the qualification that there is an existing AEL, Auto-X has an existing developed footprint, fenced off, and accessed through Lindsay and Stanford Roads. Thus, there is no construction activity associated with the line addition.

No other site access is necessary, as the production increase will utilise existing facilities. Existing internal roads will also continue to accommodate traffic.

The current site measures about 55 000 square metres (m²), of which only approximately 350 m² will be utilised for the proposed BM Line 3.

The existing operations include two BM Lines (1 and 2) and associated infrastructure. The existing infrastructure, i.e. material storage, battery case manufacturing, acid production, Pb smelting, plate manufacturing, battery charge room, product labelling and storage, will support the proposed BM Line 3. The battery assembly, purpose of this application, comprises enveloping of positive plates and interleaving these with negative plates to form elements. Elements are loaded into moulds and welded together to form cells. Cells are deposited into cases and welded together, with covers or lids heat-sealed onto the cases and end terminals burnt. The batteries are then subjected to a leak test. The battery assembly process is illustrated in **Figure 2**, while the broader battery manufacturing and assembly process including the existing supporting infrastructure is indicated in **Figure 3**.

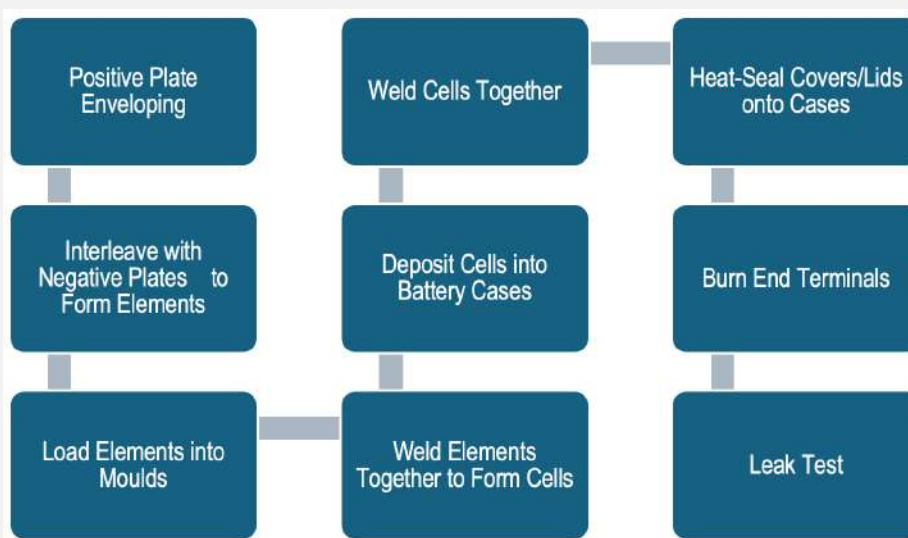


Figure 2: Battery assembly line (this involves putting components together, no manufacturing)

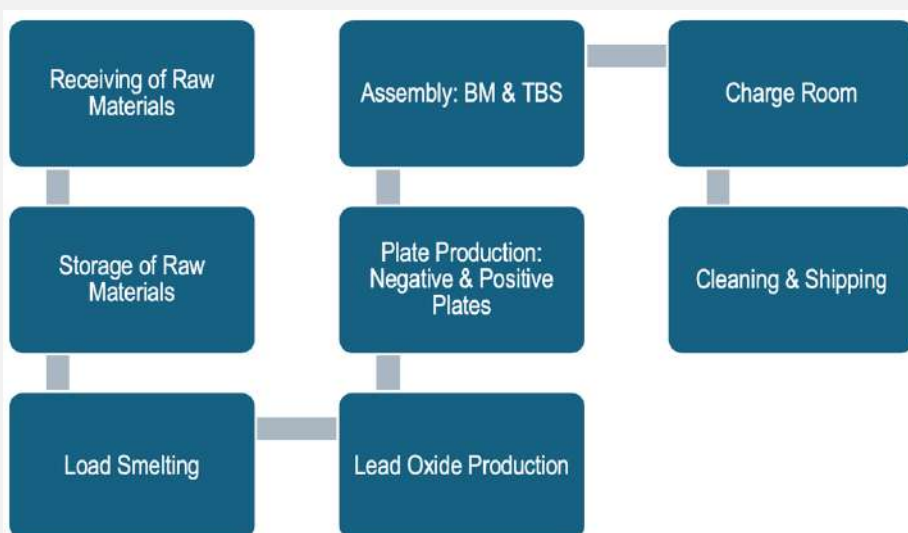


Figure 3: Broader process currently undertaken on site (already authorised, not part of this application)

The BM Line 3 operation will increase annual production by 220 000 automotive batteries to total 1 599 647 production capacity per annum, which remains within the maximum permitted consumption rates (2 533 450) specified in the AEL.

The production increase is within limits provided for in the AEL in that the proposed increase in Pb alloy consumption (2 432 393.3 kilograms per annum [kg/a]) remains within the facility's maximum permitted consumption rate (25 998 910 kg/a).

Auto-X has lodged an AEL variation application with the NMBM and EA application with the Eastern Cape Department of Economic Development, Environmental Affairs and Tourism (DEDEAT/ the Department) in line with the requirements of the NEMAQA, the Regulations, and the Act. As such, Abantu was appointed as a *Reg. EAP* to lodge the EA application and conduct a Basic Assessment (BA) process for the addition of BM Line 3.

1.2. Potable Water, Wastewater, Stormwater, and Solid Waste

The production increase is not envisaged to trigger any amendments to the existing potable water, wastewater, and stormwater infrastructure.

The existing waste management practices will continue on site, i.e. NMBM routine refuse removal service. Any hazardous wastes, i.e. industrial effluent generated by BM Line 3 will be managed through existing approved waste management systems and licensed service providers.

Water is provided by the NMBM at a fee. Two reservoirs exist on site to ascertain back-up during down periods, such that there is no interruption in operations which run throughout the year save for maintenance time. Additionally, Auto-X is a holder of a Water Use Licence (WUL), licence number 09/M20A/A/17542, for capturing 11 000 cubic metres (m³) of groundwater seepage. It is not the plan of the applicant to exceed the amount of water authorised for abstraction. As such, no WUL amendment is planned as part of the BM Line 3 operation. Furthermore, the applicant captures rainwater for use within the non-potable areas of the operations. Lastly, Auto-X operates a reverse osmosis plant to improve the quality of water for application within the site. This plant also assists in recycling of water within some areas of the operations.

1.3. Electricity

It is not anticipated that the proposed production increase will require any changes to the existing electricity infrastructure, but only the consumption amount. Auto-X has a gas tank installed on site for the containment of liquified petroleum gas (LPG), used on site to lessen electricity consumption. It is envisaged that the operation of BM Line 3 will result in the increase of LPG consumption by 5 000 kg/a. The expansion of the gas tank is not planned, but more refill trips to cover the increase in consumption.

Various measures are implemented within the existing facility as part of saving energy; these include insulated chilled water pipelines to keep the water cool such that less energy is required for cooling; use of energy efficient generators; light-emitting diodes/ LED; traditional geysers were replaced with solar heaters; etc. As such, Auto-X is a holder of an ISO 50001 Certificate issued in 2023 for energy management.

1.4. Property Ownership and Facility Management

The operations occur on Erf 2619 and 2620 of the Korsten Township, although the BM Line 3 addition will take place on Erf 2619. The properties are owned by Erf 2619 (Pty) Ltd and Erf 2620 (Pty) Ltd, both owned by the applicant. The facility is also operated by the applicant.

1.5. Receiving Environment

A Screening Report was generated from the National Web-Based Environmental Screening Tool. The findings of the Screening Report are as depicted in **Table 1** below regarding sensitivity of the site, and the specialist studies identified based on the sensitivities before verification are listed thereafter.

As per Government Notice (GN) No. 320 and 1150 of 2020 enacted in terms of sections 24(5)(a) and (h) including section 44 of the Act, prior to undertaking any specialist studies or assessments Site Sensitivity Verification (SSV) for

each theme must be undertaken to confirm or dispute the findings of the Screening Report.

Table 1: Site Sensitivity

Theme	Site sensitivity
Agriculture	High
Animal species	Medium
Aquatic biodiversity	Very high
Archaeological and cultural heritage	Low
Civil aviation	Very high
Defence	Very high
Palaeontology	Very High
Plant species	Medium
Terrestrial biodiversity	Very high

The specialist studies or assessments identified as per the Screening Report, prior to verification:

- Agricultural;
- Ambient Air Quality/ Air Quality;
- Aquatic Biodiversity including Hydrology;
- Archaeological and Cultural Heritage, including Palaeontology;
- Health;
- Hydrology;
- Landscape/ Visual;
- Noise;
- Socio-Economic;
- Traffic; and
- Terrestrial Biodiversity including Animal and Plant Species.

A SSV report was compiled for the above listed themes, refer to **Appendix G4**.

1.5.1. Climate

The weather pattern of the area changes throughout the day as it lies at the confluence of several climatic regimes, the most important of which are the temperate and subtropical regimes (Stone 1988). Maximum and minimum mean temperatures are experienced in February and July, respectively (McCallum 1981). Exceptionally elevated temperatures, approximately 30 degrees Celsius (°C), can occur during berg wind conditions that develop frequently in autumn and winter. The mean annual rainfall for the Gqeberha area is approximately 600 millimetres (mm) as per Stone (1988).

Gqeberha is dominated by topographical or gradient winds for most of the year (Grobler, 2012). The area experiences westerly winds throughout the year, although in summer the percentage of easterly winds reaches more than 40% (Schumann et al, 1999). The strongest winds occur during October and November, with weakest winds during May and June (Schumann et al, 1991).

1.5.2. Geology, Soils, and Agriculture

In geological terms, the site is predominantly underlain by Kirkwood Formation with a little patch of Peninsula, Pakhuis, Cederberg Formations (**Figure 4** below).

The Kirkwood Formation is part of the Uitenhage Group in the Eastern and Western Cape provinces, primarily within the Algoa and Gamtoos Basins. It formed during the Late Jurassic to Early Cretaceous (approximately 145 to 135 million years), along the southern margin of Gondwana (Muir et al, 2017). The formation is composed of fluvial sedimentary

rocks, including variegated mudstones, medium-grained lithic sandstones (often charcoal-rich), and sporadic conglomerates (Muir et al, 2017). It is subdivided into three members, the Swartkops (estuarine sandstones), the Colchester (fossil-rich shales, siltstones, and sandstones), and the Bethelsdorp (marine microfossils in shales and sandstones) as reported by Muir et al (2017).

The Peninsula, Pakhuis, and Cederberg Formations are part of the Cape Supergroup and are significant geological features in the Cape Peninsula (<https://vogelgat.co.za/geology/>). The Peninsula Formation Sandstone, also known as the Table Mountain Sandstone, is a key formation that has given its name to the Table Mountain Group. It consists of hard, light grey quartz arenite sandstone that is resistant to erosion (<https://vogelgat.co.za/geology/>). The Pakhuis Formation is identifiable by glacially deposited pebbles of sandstone (<https://vogelgat.co.za/geology/>). The Cederberg Formation includes dark grey siltstone and silty sandstones, which have been eroded into various structures by the wind (<https://vogelgat.co.za/geology/>).

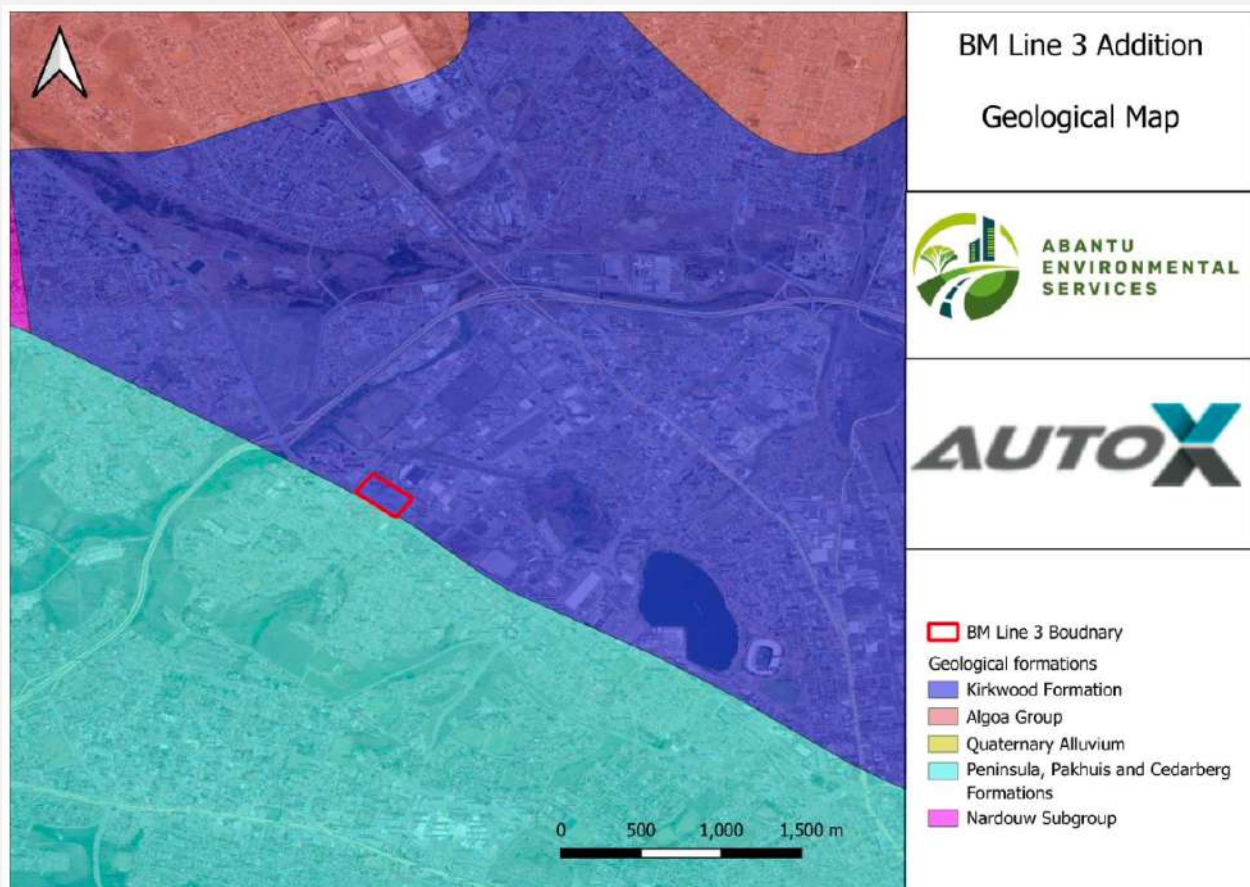


Figure 4: Geology of the proposed site

The soil class comprise imperfectly drained soils, often shallow with a plinthic horizon. The soils that prevail on site have a marked clay accumulation, are strongly structured, and have a non-reddish colour. They may occur associated with one or more of vertic, melanic, and plinthic soils. Their relative wetness is favourable in dry areas, but may be seasonally wet in opposite areas.

There are no agricultural activities taking place on site, this has not been the case in more than 30 years as per available historical imagery (Google Earth, 2026). Additionally, the site has had an existing industrial facility for several years. As such, the land on which the addition of BM Line 3 is to be undertaken is already utilised for industrial purposes. Thus, the “high” agricultural sensitivity as depicted in the Screening Report is disputed and revised to “very low to no” sensitivity with no further assessment recommended.

1.5.3. Archaeology, Cultural Heritage, and Palaeontology

As indicated in subsection 1.5.2, the site can be summarised as Late Jurassic–Early Cretaceous, fluvial sediments, extremely fossil-rich, part of the Uitenhage Group. The Kirkwood Formation is the most fossil-rich unit of its age, yielding remains of theropods, sauropods, ornithopods, stegosaurs, ankylosaurs, and basal ornithomimosaurs such as *Nqwebasaurus* (Muir et al, 2017). Notable discoveries include the juvenile dryosaurid *Iyuku raathi*, *Paranthodon*, and various reptiles, amphibians, insects, mammals, and plant fossils such as ferns, conifers, cycads, and bennettitaleans (Muir et al, 2017). Mesozoic amber has also been recovered, providing insights into ancient ecosystems (Muir et al, 2017). Although the fossil-rich fossil heritage is acknowledged, the fact is that construction will not take place on site. The “very high” palaeontology sensitivity is revised to “very low to no” sensitivity and no further assessment necessary.

The archaeological and cultural heritage sensitivity of the site, “low”, is also disputed and revised to “very low to no” sensitivity. Due to the fact that no construction will take place on site, no assessment is recommended.

1.5.4. Topography and Drainage/ Hydrology

The land is relatively flat on site, occurring at an elevation range of 40 to 47 metres (m), and slopes towards the east. The site drains by means of surface flow, primarily in an eastern direction towards the existing NMBM stormwater infrastructure. It is expected that groundwater flow mimics the surface water flow.

1.5.5. Terrestrial Biodiversity including Animal and Plant Species

The study area is located within the Albany Thicket Biome. The site falls within the Motherwell Karroid Thicket vegetation/ ecosystem type (**Figure 5**). The vegetation/ ecosystem type is classified as Critically Endangered (GN No. 2747, 2022). Approximately 9(+3)% of the vegetation type/ ecosystem is protected, with the conservation target at 19% (South African National Biodiversity Institute [SANBI], 2018).

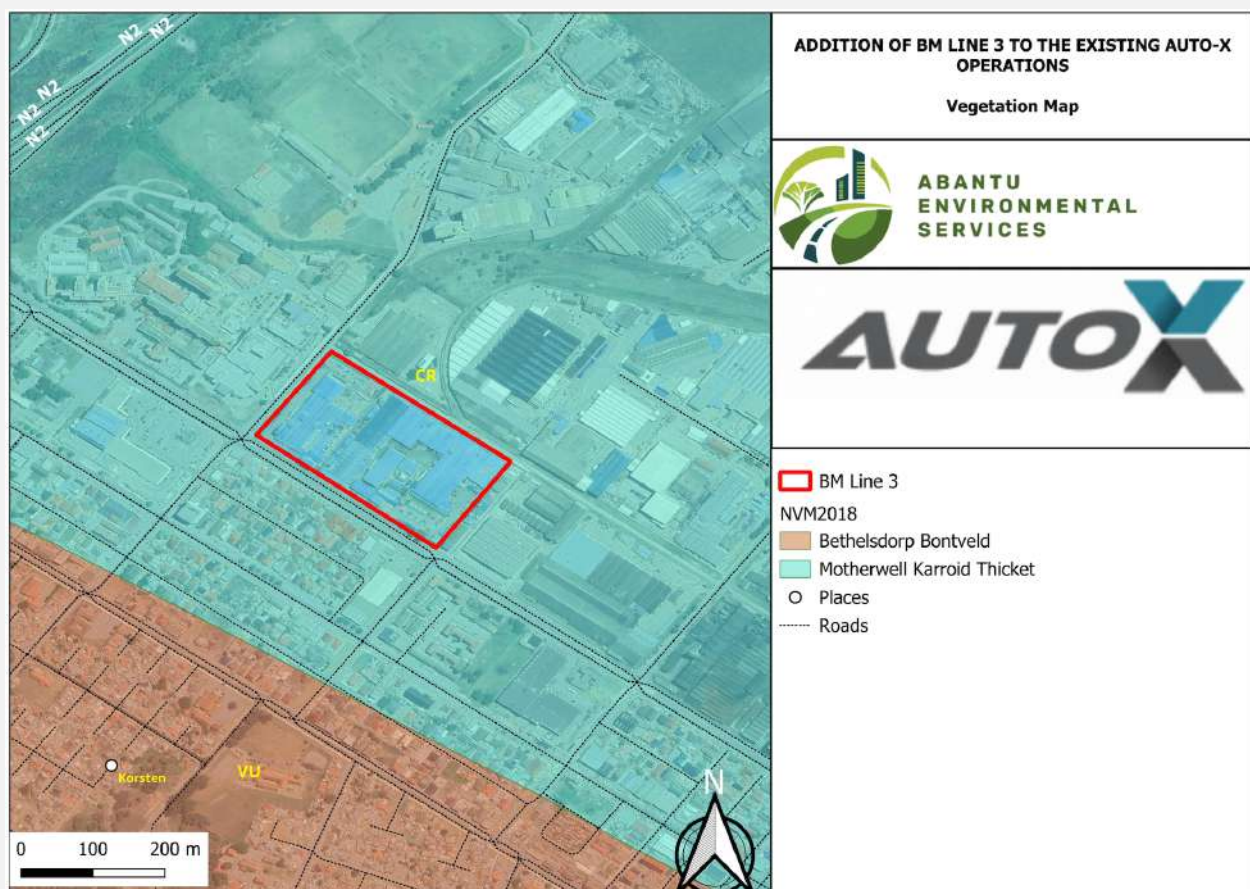


Figure 5: Vegetation type that would prevail on site if it were vegetated

It should be noted that no vegetation, including animal species, occur on site and, even if they did, the proposed BM

Line 3 addition does not include any construction but installation and operation of machinery on the existing warehouse currently utilised for similar activities.

As indicated in **Figure 6** the site does not occur within a terrestrial Protected Area (PA), Critical Biodiversity Area (CBA), Ecological Support Area (ESA)/ Critical Ecological Support Area (CESA), and any other sensitive area (Eastern Cape Biodiversity Conservation Plan [ECBCP], 2019; NMBM, 2009).

The “very high” terrestrial biodiversity, “medium” animal and plant species sensitivities are therefore disputed. The ecological sensitivity for terrestrial biodiversity (including flora and fauna) is revised to “very low to no” sensitivity and no further assessment is recommended more so as no construction is associated with BM Line 3.



Figure 6: Biodiversity sensitivity

1.5.6. Aquatic Biodiversity

The site falls within the Mzimvubu to Tsitsikamma Water Management Area. At a desktop level, the site constitutes aquatic ESA 1, refer to **Figure 7** below (SANBI, n.d; ECBCP, 2019). The ESAs are vital regions that support the functioning of PAs and CBAs, playing a crucial role in delivering ecosystem services and enhancing landscape connectivity (SANBI, 2018). The ESAs are defined as areas in a natural, near natural, or moderately degraded condition that are not essential for meeting biodiversity targets but are important for supporting the functioning of the areas referred to above. They are often vital for delivering ecosystem services, which include benefits such as clean water, climate regulation, and habitat for wildlife (SANBI, 2018). As per the site investigation conducted in June 2026, the site is in a developed state and no longer displays the sensitivity referred to.

Rivers, Papkuils and Papenkuils, around the general area occur more than 175 m from the site. The National Freshwater Ecosystem Priority Area (NFEPA) wetlands are located more than 750 m from the site.

The majority of the site extent has hardened surfaces which drain to the existing stormwater infrastructure during rainfall

events.

Thus, the “very high” aquatic sensitivity associated with the site due to it being categorised as ESA 1 on a desktop level is disputed. The sensitivity is revised to “very low to no” sensitivity, and the recommendation is that no further assessment is warranted in this regard. This is amplified by the fact that no development will take place on site.

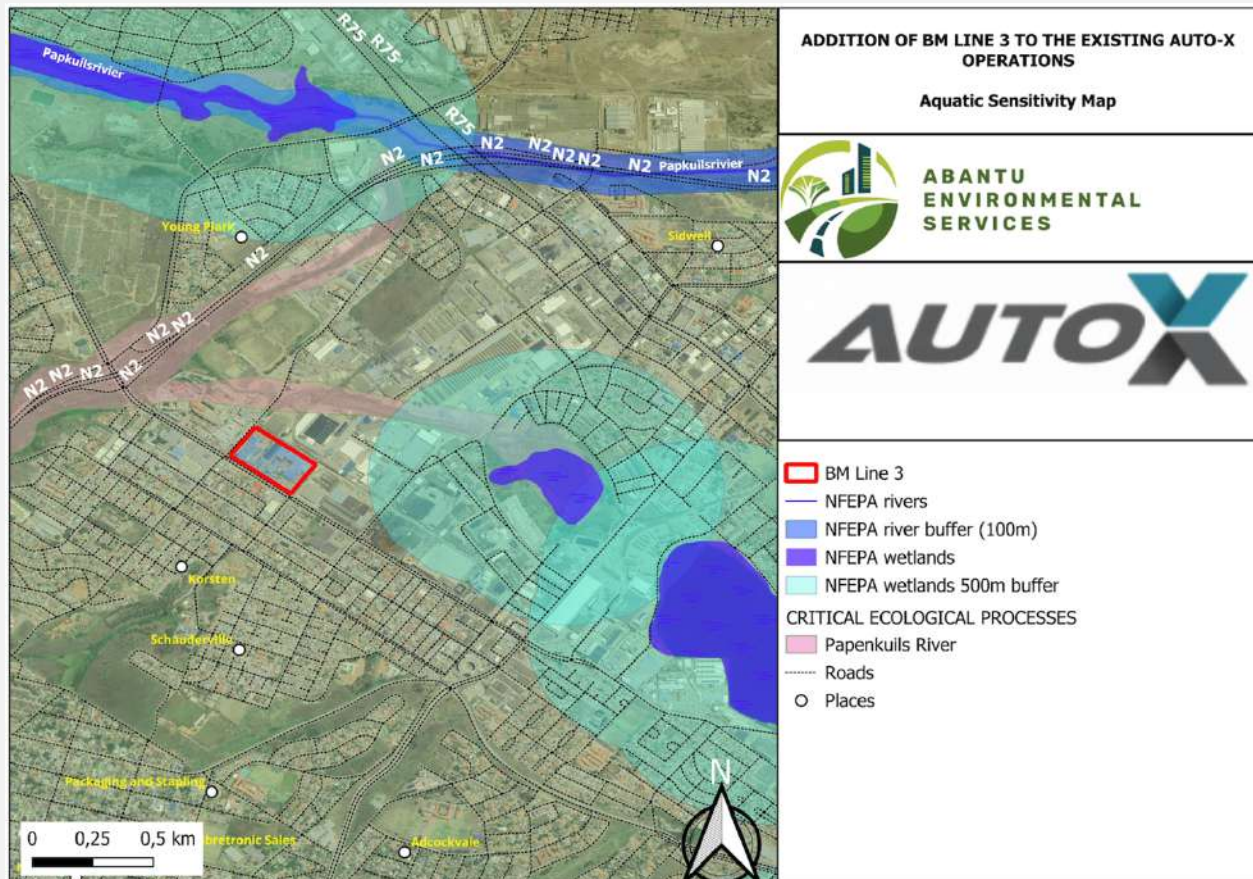


Figure 7: Aquatic sensitivity associated with the site

1.5.7. Civil Aviation

The site presents a “very high” civil aviation sensitivity as per the Screening Report, this means a site situated within 8 and 15 kilometres (km) from a major civil aviation aerodrome. In fact, the Chief Dawid Stuurman International Airport is located within 7 km (in a straight line) of the site.

It should be noted, however, that the facility has operated on site without any reported or recorded impacts to civil aviation. As such, it is not anticipated that the operation of BM Line 3 will pose any impact to the civil aviation. Thus, the “very high” sensitivity is disputed and revised to “very low to no” sensitivity. No further assessment is necessary.

1.5.8. Defence

The Screening Report classifies the defence theme sensitivity of the site as “very high”. However, there are no known military bases/ facilities present within the vicinity of the site. The nearest military facilities to the site are located in Forest Hill and Summerstrand, more than 7 km (in a straight line) from the study area. As such, it is not anticipated that any impacts will emanate from the site to affect this theme. Additionally, the operations are already existing on site and there has been no report of any effect to this theme. As such, the sensitivity is revised to “very low to no” sensitivity and no further assessment required.

1.5.9. Socio-Economic Aspects

The site is located in NMBM, one of eight Metropolitan Municipalities in South Africa. In 2001, the NMBM was

formed as a single administrative area covering inter alia Gqeberha, Kariega, Despatch, and several surrounding areas.

The city has a unique advantage in that it possesses two ports, namely Port Elizabeth Harbour and Port of Ngqura. This presents several opportunities related to the diversification of port activities, the maritime sector, and the distinction between 'dirty' and 'clean' port uses. The NMBM covers an extent of 195 890.76 hectares (Almond et al, 2008).

1.5.9.1. Population

The current population of NMBM is estimated at 1 190 496 (StatsSA, 2022), with a growth rate of 1.54% (IHS, 2017), which is lower than that of other metropolitan areas in South Africa, such as Ekurhuleni (2.1%) and Tshwane (2.6%). According to StatsSA (2022), the NMBM population is split 52.7% to 47.3% in favour of females, with the African population accounting for 62.7% of the overall population, Coloureds accounting for 19.2%, Whites 15.7% and the rest are a mix of different ethnicities. As per StatsSA (2022), the elderly constitutes 8.8% of the population while the young children under the 14 years make up 22.3% of the population. The literacy level in the local municipality, by 2016 were measured at 76% of the population, which is a significant increase from the 66% measured in 2006.

1.5.9.2. Education

According to the municipality profile (NMBM, 2016) there is a total number of 324 schools, which consist of 269 ordinary public schools, 11 special needs schools, 25 independent schools and 19 Early Childhood Development Centres within the municipality. A comparison with other metros shows that 75.8% of learners aged 7 to 24 years in the municipality attended an educational institution as compared to 80.2% in Buffalo City and 79.2% in Mangaung (79.2%). The lowest learner attendance was observed in Cape Town (69.8%) and eThekweni (71.1%). Persons 20 years and older with less than Grade 7 as highest level of education are 12.8% (StatsSA, 2022).

1.5.9.3. Economic Outlook

Whilst the NMBM is undoubtedly an important node of activity within the economy of the Eastern Cape, it is characterised by several challenges in terms of economic development. These include:

- A high unemployment rate (24.5%), according to the NMBM Integrated Development Plan (IDP) of (2022/23-2026/27), compounded by the low education levels of the labour force, including large numbers of illiterate adults with limited employment prospects.
- Ageing and inadequate investment in the maintenance and upgrading of infrastructure.
- The dependence on the automotive sector and insufficient diversification within the manufacturing and others.
- A lack of up-to-date local economic statistics, monitoring, and evaluation systems.

As per the Eastern Cape Socio Economic Consultative Council, the Gross Domestic Product (GDP) growth rate for the NMBM was 2.1% in 2010 and the GDP per capita R 52 147. The largest economic sectors in the NMBM are manufacturing, finance, community services, and transport. Community services, trade, and manufacturing sectors are the sectors that create the most employment in the NMBM. Good quality infrastructure is key to sustainable social, economic, and industrial development. Poor infrastructure hampers development, growth, and ability to trade in the domestic and global economy.

Economic infrastructure, which includes transport, energy, telecommunications, water, and sanitation provides services which are of fundamental importance for development. In NMBM it is deficient, and this is exacerbated by inadequate maintenance and thus prematurely deteriorating installations and services. Infrastructural services are often overlooked as a means to alleviate poverty and improve environmental conditions. In order for NMBM to grow and develop a sustainable economy, it is important to build new economic drivers to replace or augment the ones that have served the region in the past.

According to the NMBM IDP (2022/23-2026/27), 45% of households in NMBM receive at least one social grant as a

source of income. A total of 58% of the population depend on the capital grant dependency. A Socio-Economic Assessment is thus not recommended.

1.5.10. Landscape/ Visual

The site is already an operational battery manufacturing and assembly facility. The operation of BM Line 3 will not have any more effect than currently is the case to the sense of place in the general area. As such, the landscape/ visual assessment is not warranted.

1.5.11. Noise

Noise in battery assembly lines can be caused by various factors, including the operation of automated assembly systems, robotic welding machines, conveyor lines, and inspection stations. These machines, when operating simultaneously, can lead to high sound levels.

Mitigation measures, i.e. noise absorbers, are in place throughout the facility to absorb sound energy, improving overall comfort in the battery assembly environment. Additionally, it was noted during the site investigation, that the workers around noise generating areas are equipped with noise ear blockers. It is the understanding of the *Reg. EAP* that this practice will continue on site. Due to the fact that the site is already in operation, and this will be a continuation of the operations, the noise assessment is not recommended.

1.5.12. Traffic

As indicated above, the site is accessed from Lindsay and Stanford Road. However, observation was that trucks fetching completed battery batches access the site via Stanford Road only. Trucks were indeed observed parked around the site, along Stanford Road. These were neatly parked such that there was no traffic obstruction on this road. Based on the evacuation of batteries, batch evacuation on completion, truck arrival is known ahead of time and is done only when required. Thus, this practice will continue to be observed as part of the operations on site. Therefore, a traffic assessment is not recommended.

1.5.13. Health

The health impacts of lead-acid battery assembly are multifaceted. Workers in battery assembly are at potential risk due to exposure to Pb, where appropriate personal protective equipment (PPE) is not worn/ PPE is not worn correctly. Lead can be absorbed into the body by inhalation and ingestion, leading to various health issues such as neurological damage, kidney damage, and reproductive issues. Lead exposure can also cause anaemia, miscarriages, and damage to organs responsible for sperm production.

Lead-acid batteries are also a source of environmental pollution, releasing Pb and sulfuric acids (H_2SO_4) into the environment, which can impact air, soil, and water quality. The improper disposal of lead-acid batteries can lead to harmful substances leaking into the soil and groundwater, posing risks to ecosystems and communities relying on local water sources.

To mitigate these health risks, Auto-X currently follows crucial safety measures, handles lead-acid batteries with care, and disposes of them responsibly. This practice will continue as part of the BM Line 3 addition. Recycling processes are also practiced, reducing health risks and environmental impact. The operator/ applicant implements and will continue to implement stringent safety protocols, including proper ventilation, employee training, and the use of appropriate PPE, i.e. masks (**Figure 8** below). Regular monitoring of air quality and adherence to safety protocols are some of the current essentials for ensuring the safety and health of workers in this facility. Auto-X is a holder of an ISO 45001: 2018 (Occupation Health and Safety Management System). Thus, a health assessment is not warranted.

1.5.14. Ambient Air Quality/ Air Quality

The potential air quality impacts related to battery assembly lines are problematic due to the production of airborne contaminants that can pose health risks to workers and compromise product quality. These contaminants can

contaminate the production environment, affecting battery performance, consistency, equipment reliability, and fire safety. During the site investigation, the operations were neat and free of any visible contaminants.

Process emissions across the plant are captured at source by local extraction and filtration systems and vented to atmosphere through 40 stacks (including the addition of AX4 as part of BM Line 3) ranging from 4 to 8.5 m in height. Although this is not part of the BM Line 3, it is worth mentioning that the stacks servicing the lead-oxide (PbO), plate production, and assembly activities including the battery acid filling and charge room carry emissions limits that must be complied with by Auto-X. In compliance with the emission limits, Auto-X has fitted filters/ cartridges, baghouses, and/or wet scrubbers throughout the facility (**Figure 8**) and routinely commissions air quality specialists to conduct stack emissions sampling to understand whether the emissions are still within limits. The recent stack emissions sampling was undertaken in late July to early August 2025 by Yellow Tree. The findings are discussed below.



Figure 8: Existing post-classifier bag filter (upper left), ambient extraction (upper centre), wet scrubber (upper right), BM assembly line extraction point (lower left), and stack (lower centre and right)

Other measures implemented on site to ascertain cleaner production targets include improved stack sampling accuracy, tracking of absolute emissions (target reductions), air dispersion modelling investigations, routine duct damage inspections and repairs, cleaning and maintenance of abatement equipment, audits on ducting (assessment of ducting to ensure flowrates are sufficient for extracting emissions from workstations), welded windows (welded windows ensure that all of the emissions are extracted through extraction systems that feed into the abatement equipment), etc. Masks are worn at all times inside the facility; this was observed during the site investigation.

Efficient fume extraction systems are essential for maintaining clean production environments, protecting workers, and ensuring product reliability. These systems capture airborne particles, gases, and toxins, reducing exposure and contamination risk during battery assembly. These also support environmental emission standards and safer

manufacturing practices. Furthermore, back-up plans employed within the facility include temporary cessation of operations where abatement measures do not deploy as normal.

Lastly, an Environmental Management System (EMS) that gives effect to the principle of continuous improvement was established by Auto-X; this EMS identifies and quantifies potential for environmental impacts, prioritises the identified impacts, identifies appropriate preventative and corrective actions, lays out developed responsive management controls, systems, and procedures, and identifies improvement projects to be added to the five-year Environmental Improvement Programme. Auto-X is a holder of an ISO 1401:2015 certificate (EMS). These management and mitigation measures will continue for the lifetime of the facility.

1.5.14.1. Wet Scrubbers

The information provided in this section is extracted from the Stack Emissions Sampling conducted by Yellow Tree in 2025. Wet scrubbers (example in the figure below) are a commonly used abatement solution that employs absorption as the primary mechanism to remove particulate matter (PM), sulphur dioxide (SO_2), and sulphuric acid (H_2SO_4) mist. Wet scrubbers that use solely water as a scrubbing liquor rely primarily on the physical impact (impingement) of water droplets on PM; and rely on the solubility of gaseous components in water. In a spray-tower scrubber, the particulate-laden flue gas enters through the bottom of the scrubber, and clean scrubber liquor is sprayed through nozzles that are at the top of the scrubber or distributed throughout the scrubber. A liquor purge and fresh supply are necessary to enable continuous removal from the gas stream before equilibrium concentrations are reached. Water particles impact upon, and entrain the PM, thereby removing particles from the gas stream. Wet scrubbers may have various designs, including venturi, spray nozzle, orifice, and mechanical scrubbers. These technologies employ different mechanisms by which to increase the mixing of the gas and droplets for increased removal efficiency.

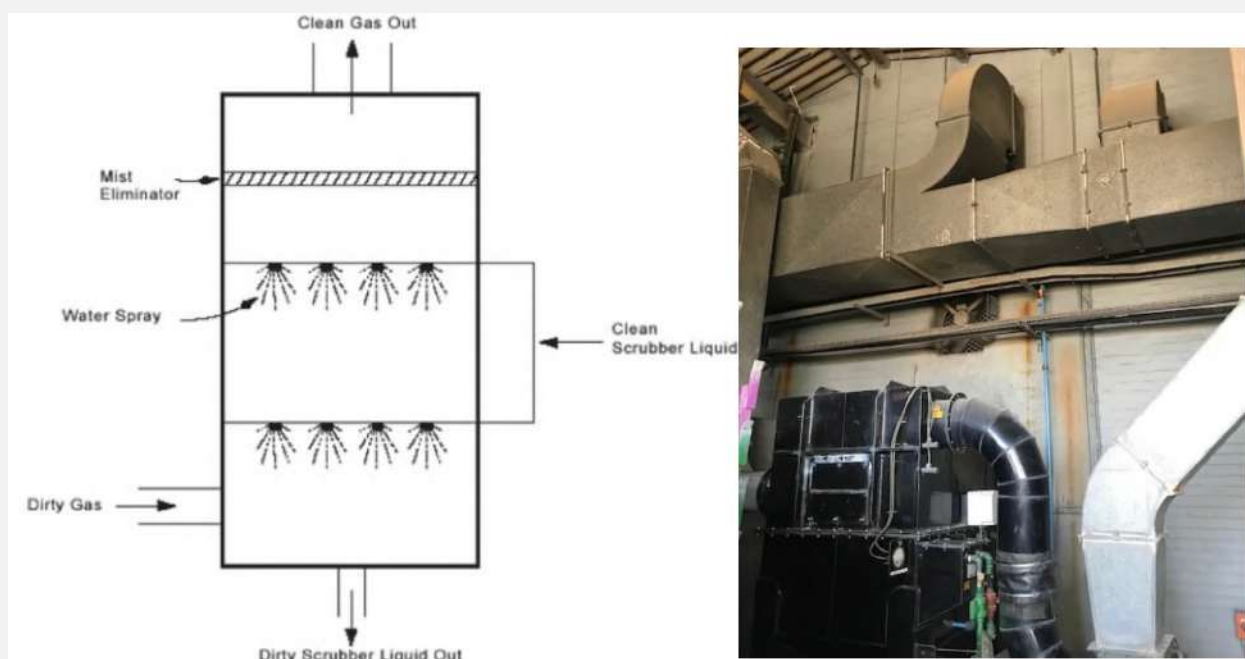


Figure 9: Diagram of a simple wet scrubber (left) and one of wet scrubbers currently employed on site (right)

1.5.14.2. Bag Filters

The information provided in this section is a continuation of extraction from the 2025 Stack Emissions Sampling. Bag filters typically reduce PM emissions to a level near 15 milligram per cubic nanometre (mg/Nm^3), and the use of a high-efficiency particulate air (HEPA) filter means that even lower PM emissions than $15 \text{ mg}/\text{Nm}^3$ can be removed. Because PbO is expected to be in particulate form, it was expected that most of the Pb would also be removed by the bag filters and HEPA filters.

Reverse pulse (also known as reverse jet) bag filters are the most common bag filter. A bag filter may also be known

as a baghouse. Inside reverse pulse bag filters, the filter bag forms a sleeve supported by a cylindrical frame. Process/ flue gas containing high concentrations of PM flows into the main housing below the tube sheet. Any PM is collected on the outside of the filter bag while the clean air passes through the filter bags and exits above the tube sheet. When sufficient PM has been collected on the bags so that the pressure drop across the bags exceeds the setpoint, bag cleaning takes place. Bag cleaning is performed by jets of compressed air which dislodge PM caked on the outside of the bags. The jets are of a sufficiently high pressure that process/ flue gas flowing into the bag filter unit does not have to be stopped during cleaning. A diagram of a reverse pulse bag filter is shown in **Figure 10**. A number of the bag filters, such as those serving the process at Auto-X, are Donaldson bag filters that have filter elements positioned at an angle, for ease of access, instead of vertical filters.

Abrasion or tearing of filters is a common cause of poor performance from filter-type abatement equipment. Satisfactory abatement efficiency can be maintained by ensuring that inspections of the filters and filter bags are built into a maintenance plan to prevent tearing and abrasion.

HEPA filters are a type of mechanical air filter that can remove at least 99.97% of dust, bacteria, and other airborne particles with a size of 0.3 microns.

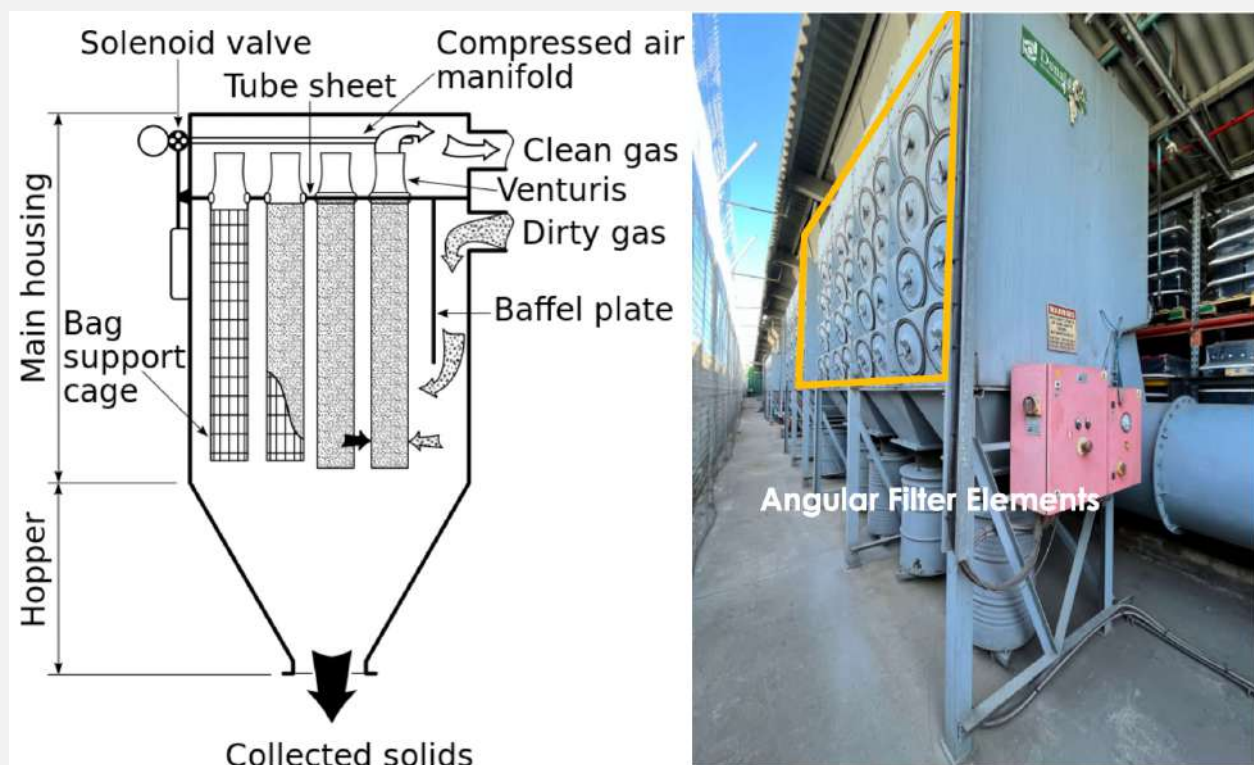


Figure 10: Diagram of a reverse pulse bag filter (left) and a current bag filter at the site (right)

1.5.14.3. Findings of the 2025 Stack Emissions Sampling

As per the 2025 Stack Emissions Sampling, there were no PM emissions exceedances (30 mg/Nm^3) from the stacks servicing the PbO Production Area. Stacks AX12 and AX13 serve the post-classifier bag filters. The purpose of these stacks is to extract the air that is used as a transport medium for conveying the PbO powder from the reaction pots to the oxide blower. Stacks AX14 and AX15 serve the oxide mill silos. The purpose of these stacks is to extract the air that is used as a transport medium for conveying the fine PbO powder to the silos. Stacks AX12 to 15 are served by baghouses in combination with HEPA filters and met the respective emissions limits for PM and Pb in 2025. Stack AX10 serves general extraction fans in the PbO manufacturing area and is equipped with a wet scrubber.

According to the 2025 Stack Emissions Sampling, there were no Pb emissions exceedances for some of the stacks servicing the PbO Production Area, i.e. AX13 to 15. The Pb emissions from AX10 were 1.81 mg/Nm^3 , although there is

a measurement uncertainty of 0.44 mg/Nm³, which means that the true measurement could have been slightly above or below the emissions limit (2 mg/Nm³). All scientific measurements contain a degree of uncertainty, and the ISO 17025 quality management system requires that this uncertainty be quantified and reported. The Pb emissions from AX12 had exceeded the limit. The exceedance of Pb from AX12 may have indicated that the bag filters, or HEPA filters, serving this stack should have been inspected for tears or abrasion that may impact their effectiveness. AX12 was identified as a priority stack following the 2025 sampling campaign, Auto-X immediately replaced the HEPA filters. This demonstrates Auto-X's commitment to continuous improvement and rapid corrective action.

There were no emissions exceedances from any of the stacks in the Plate Production and Assembly Area. Stacks AX6 to 8 and 22 are each equipped with a bag filter as abatement equipment. Bag filters typically reduce PM emissions to a level near or below 15 mg/Nm³, and so the low PM emissions were anticipated. Stacks AX1, 9, and 16 are each served by a wet scrubber as abatement equipment, and these scrubbers are working well to reduce PM and Pb emissions to a level below the limit. Stack AX17 serves one of the six electric ovens, while stack AX18 serves one of the eight indirectly fired gas-fired ovens. These stacks are not served by any abatement equipment, and PM and Pb emissions from these stacks were below the limit. Yellow Tree sampled Total Volatile Organic Compounds (TVOCs) from AX19, as the lid sealing operation involves plastic welding that could release TVOCs. There were no emissions limit for TVOCs at the time of compiling the 2025 Stack Emissions Sampling report and negligible TVOCs emissions were measured, indicating that the plastic welding was not releasing EPA 25(b) detectable TVOCs.

The emissions that were measured from the Battery Acid Filling and Charge Room Area stacks, i.e. AX2 to 3 and 20, did not exceed emission limits. There are four vents that serve the battery acid filling line, and Yellow Tree sampled one of these vents (AX20). Average SO₂, H₂SO₄ mist/ sulphur trioxide (SO₃), and nitrogen oxides (NO_x) emissions from AX2 to 3 were below the respective limits. A wet scrubber serves each of the stacks. Average SO₂, H₂SO₄ mist/ SO₃, and NO_x emissions from AX20 were below the respective limits. The stack is not served by any abatement equipment. No nitric acid, or other major sources of NO_x emissions were present in the Battery Acid Filling or Charge Room Area, and so negligible NO_x emissions were expected.

In summary, the 2025 Stack Emissions Sampling observations, taking into consideration the 2023 and 2024 Stack Emissions Sampling, were:

- The Pb emissions from stacks AX1, 6 to 9, 16, and 22 remained negligible during all three sampling campaigns (2023, 2024, and 2025).
- Emissions from AX10 had increased during the last three sampling campaigns, which may indicate that the scrubber performance has dropped. However, the scrubber is working well to reduce the emissions to a level below the limit. The increase in emissions may also indicate that in 2025, more fugitive PbO dust was present in the oxide production area from which AX10 expels emissions. Auto-X cleans the area surrounding the PbO reactor pots every two weeks, thus, some variance in fugitive PbO may be expected. Although the limits were not exceeded, mitigation measures such as enhanced housekeeping, regular duct inspections, and maintenance of the wet scrubber were implemented within the site.
- The Pb emissions from AX12 had exceeded the limit of 2 mg/Nm³, which indicated that the bags within the bag filter or HEPA filters may have need replacement, as previously discussed. As indicated above, this was addressed by the applicant as soon as the report was in receipt of Auto-X.
- The Pb emissions from AX13 had increased by 35.9% from 2024 to 2025. Despite the increase, emissions remained below the limit. The stack is served by a baghouse in combination with HEPA filter which is working well to reduce emissions to a level below the limit. Routine maintenance and inspection of the baghouse and HEPA filters continued/ continues as per the scheduled preventative maintenance programme.
- The Pb emissions from AX14 had decreased by 126.5% from 2024 to 2025. The stack is served by a baghouse in combination with high efficiency.
- The AX15 stack did exceed the limit in 2024. However, emissions decreased to a negligible level during the 2025

sampling campaign. The stack is served by a baghouse in combination with HEPA filter which is working well to reduce emissions to a level below the limit.

- The Pb emissions from AX17 and AX19 had increased from 2024 to 2025, but these stacks are not equipped with abatement equipment and so emissions may vary depending on the plant operation at the time of sampling.
- The Pb emissions from AX18 had decreased by 132.4% from 2024 to 2025.

1.5.14.4. Findings of the 2026 Air Quality Impact Assessment

The information provided in this section is extracted from the Air Quality Assessment conducted by Lethabo Air Quality Specialists (LAQS) in 2026. Three scenarios were modelled by LAQS, i.e. Scenario A: site only, with emissions derived from the flow rates given in the AEL application (AEL schedule permit) form and the maximum permitted emission concentration applicable to each stack. Scenario A represents a theoretical 'worst-case' assessment based on the maximum emission limits in the AEL. It is not a prediction of actual plant emissions; Scenario B: site only, using the emissions the plant is expected to release based on the 2025 measured emission rates and using estimated emissions for the proposed expansion based on similar process measurements; and Scenario C: cumulative impact, using the site emissions as per Scenario B together with actual emissions from surrounding industrial sources. Thus, Scenario B is based on a stack-testing campaign for the plant and typical (expected) emissions from the new stacks (BM Line 3). Therefore, the below discusses the findings for Scenario B.

The maximum predicted 24-hour 99th-percentile PM₁₀ concentration was 12.2 micrograms per cubic metre (µg/m³) located 240 m south-east of the site. This was well below the 75 µg/m³ contemplated in the National Ambient Air Quality Standards, published under Government Notice 1210 of 2009, (NAAQS). The maximum predicted annual average was 5.2 µg/m³, located 150 m east of the site centre. This was also well below the 40 µg/m³ described in the NAAQS. The maximum predicted annual average Pb concentration was 1.05 µg/m³ occurring approximately 250 m east of south-east of the site centre. This exceeded the 0.5 µg/m³ prescribed in the NAAQS. However, this exceedance was confined to the Auto-X property boundary or immediately beyond it. No sensitive receptor (school, hospital, residential area) was affected. The maximum concentration at the nearest sensitive receptor was 0.06 µg/m³, well below the 0.5 µg/m³ standard. "The Pb was the only pollutant for which a standard was exceeded on this basis. At the sensitive receptors, the highest predicted 24-hour 99th-percentile PM₁₀ concentration was 2.8 µg/m³ and the highest annual average PM₁₀ concentration was 0.5 µg/m³, both well below the NAAQS. Regarding Pb, the highest value recorded at a sensitive receptor was 0.06 µg/m³ at the Frank Joubert Primary School which was well below the NAAQS.

The impact of the operations on the ground level concentrations of SO₂ and nitrogen dioxide (NO₂) was practically negligible as per LAQS (2026). The maximum predicted 1-hour 99th-percentile SO₂ and NO₂ concentrations were 7.6 µg/m³ and 1.4 µg/m³, respectively. This was well below the applicable 350 and 200 µg/m³ illustrated in the NAAQS. In terms of SO₂, the maximum 24-hour 99th-percentile was 4.2 µg/m³ which was well below the 125 µg/m³ pronounced in the NAAQS. The maximum predicted annual average for SO₂ and NO₂ was 1.1 and 0.2 µg/m³, respectively. This was well below the corresponding NAAQS values of 50 and 40 µg/m³. At all sensitive receptors the values were even lower than these maximums.

There were no South African NAAQS for sulphuric acid at the time of compilation of the LAQS report. However, the occupational threshold limit value (TLV-TWA) applied in workplace health applications is 200 µg/m³, being the concentration to which a healthy adult worker may be exposed for eight hours per day. The occupational limit is quoted for context only and is not an ambient criterion. Regarding the assessment of ambient exposure, the Office of Environmental Health Hazard Assessment of the California Environmental Protection Agency Reference Exposure Levels (REL) were applied. A chronic REL of 1 µg/m³ as an annual average and an acute REL of 120 µg/m³ as a 1-hour average, both derived for the protection of the general population against respiratory effects applies. The maximum predicted annual average concentration was 0.2 µg/m³, occurring approximately 230 m east of north-east of the site centre and the highest value at a sensitive receptor was 0.01 µg/m³. The maximum predicted 1-hour concentration was 3.4 µg/m³. Neither reference exposure level was exceeded at the point of maximum impact or at any receptor.

On every scenario associated with the operations, the predicted maxima occur between 10 m and 260 m from the site centre. That is either on or immediately beyond the plant boundary where the short stacks and the building wake bring the plumes to ground. The nearest sensitive receptor is more than 350 m away from the site. The concentrations predicted at the receptors were therefore substantially below the maxima.

1.6. Legal Requirements

This part details the applicable legal provisions and the policy context for the BA process. It provides a review of relevant legislation, regulations, and policy documents which are applicable to or have implications for the proposed BM Line 3. The authorisation process associated with this project is carried out in compliance with South Africa's environmental legislation. The legal framework applicable to this project is diverse, a summary of the key environmental legislation and relevant policies and/or guidelines is provided in the following sections.

One of the focus points of this section is on the provisions of the Act and the NEMAQA. The Act is the primary South African legislation governing the requirements for impact assessment, while the NEMAQA governs air quality management. In the context of the proposed BM Line 3, the provisions of the Act and associated Regulations including the NEMAQA and Government Notice No. 893 of 2013, as amended (Listed Activities and associated Minimum Emission Standards) are of fundamental relevance.

This subsection, and others to follow, also describe other legislation relevant to constitutional and administrative legal precepts in South African law as well as environmental legislation of specific relevance to air quality management.

1.6.1. National Environmental Management Act 107 of 1998

The Act is the framework environmental legislation governing the requirements for impact assessment. The stated purpose of the Act is, amongst other things, "to provide for co-operative environmental governance by establishing principles for decision-making on matters affecting the environment, institutions that will promote cooperative governance, and procedures for co-ordinating environmental functions exercised by organs of state". It establishes a set of 18 principles which apply throughout the Republic of South Africa (RSA) to the actions of all organs of state that may significantly affect the environment. The Act also contains provisions on the creation of environmental management and implementation plans, and stipulates the respective organs of state responsible for doing so, as well as the content to be incorporated in such management and implementation plans.

In the context of the proposed BM Line 3, the provisions of the Act and the associated Regulations have reference. Chapter 5 of the NEMA, Integrated Environmental Management, establishes the impact assessment regime in the RSA. Since 3rd July 2006, the procedural and substantive requirements for undertaking impact assessments in South Africa have been regulated in terms of the provisions contained in section 24 of the Act and the Regulations.

The Regulations prescribe the procedural and substantive requirements for the undertaking of impact assessments and the issue of authorisations. Activities identified in terms of section 24(2)(a) and (d) of the Act, which may not commence without authorisation from the CA and in respect of which the investigation, assessment, and communication of the potential impact of such activities must thus follow the procedure as described in the Regulations. The Regulations identify lists of activities which have the potential to result in detrimental environmental impact and thus require authorisation, subject to either BA or Scoping and Environmental Impact Assessment (S&EIA).

The listed activities, in terms of Listing Notice 1, are triggered by the proposed BM Line 3 addition and, therefore, the applicable impact assessment process followed is the BA. **Table 2** provides the listed activities that are deemed applicable to the proposed project.

Table 2: Triggered listed activities in terms of Listing Notice 1 of the Regulations

Activity Description	Relevance
GNR 327 (Listing Notice 1)	
Activity 34: <i>The expansion of existing facilities or infrastructure for any process or activity where such expansion will result in the need for a permit or licence or an amended permit or licence in terms of national or provincial legislation governing the release of emissions, effluent or pollution...</i>	The Applicant proposes expansion of the operations, which includes consumption of more materials and output of more emissions albeit below the thresholds stipulated in the AEL. No modification, extension, alteration or upgrading of any facility, structure or infrastructure will take place but installation of latest battery assembly machinery.
Activity 51: <i>The expansion of facilities for the storage, or storage and handling, of a dangerous good, where the capacity of such storage facility will be expanded by more than 80 cubic metres.</i>	Liquefied petroleum gas consumption is estimated to increase by 5 000 kg/a, albeit no storage expansion is required in this regard.

1.6.2. National Environmental Management Air Quality Act 39 of 2004

The purpose of the NEMAQA is to reform South Africa's air-quality law by introducing measures to prevent pollution, protect the environment, and ensure sustainable development through national norms and standards for air-quality monitoring, management, and control.

The NEMAQA aims to protect the environment by preventing air pollution and ecological degradation; secure ecologically sustainable development while balancing economic and social development; establish national norms and standards for air-quality monitoring, management, and control across all spheres of government; and introduce specific air-quality measures to improve ambient air quality.

The NEMAQA's preamble further emphasises the need to address poor ambient air quality in many parts of South Africa; the disproportionate health burden on poorer communities caused by polluted air; the environmental and economic costs of air pollution, often not borne by the polluter; the harmful effects of ozone-depleting substances and greenhouse gases (GHG); and the constitutional right to an environment not harmful to health or well-being.

The NEMAQA also provides for a national air-quality framework guiding all levels of government; ambient air-quality and emission standards at national, provincial, and local levels; air-quality management plans integrated into broader environmental planning; and licensing of listed activities that may impact air quality.

Auto-X is already licenced to conduct activities listed in subcategory 4.13 and 7.2 of the Listed Activities and associated Minimum Emission Standards (**Table 3**). The proposed BM Line 3 does not trigger any additional activities save for the variation of the existing AEL to align with the proposed increase in material consumption, i.e. Pb and acids.

Table 3: Authorised activities in terms of the Listed Activities and associated Minimum Emission Standards

Category and Sub-Category of Listed Activity	Listed Activity Name	Description of the Listed Activity
4: 4.13	Pb smelting	The processing of Pb by the application of heat and the production of electric batteries containing Pb.
7: 7.2	Production of acids	The production, bulk handling and or use in manufacturing of hydrofluoric, hydrochloric, nitric and sulphuric acid (including oleum) in concentration exceeding 10%. Processes in which oxides of sulphur are emitted through the production of acid sulphites of alkalis or alkaline earths or through the production of liquid sulphur or sulphurous acid. Secondary production of hydrochloric acid through regeneration.

2. FEASIBLE AND REASONABLE ALTERNATIVES

“alternatives”, in relation to a proposed activity, means different means of meeting the general purpose and requirements of the activity, which may include alternatives to:

- (a) the property on which or location where it is proposed to undertake the activity;
- (b) the type of activity to be undertaken;
- (c) the design or layout of the activity;
- (d) the technology to be used in the activity;
- (e) the operational aspects of the activity; and
- (f) the option of not implementing the activity.

Describe alternatives that are considered in this application. Alternatives should include a consideration of all possible means by which the purpose and need of the proposed activity could be accomplished in the specific instance taking account of the interest of the applicant in the activity. The no-go alternative must in all cases be included in the assessment phase as the baseline against which the impacts of the other alternatives are assessed. The determination of whether site or activity (including different processes etc.) or both is appropriate needs to be informed by the specific circumstances of the activity and its environment. After receipt of this report the CA may also request the applicant to assess additional alternatives that could possibly accomplish the purpose and need of the proposed activity if it is clear that realistic alternatives have not been considered to a reasonable extent.

Paragraphs 3 – 13 below should be completed for each alternative.

3. ACTIVITY POSITION

Indicate the position of the activity using the latitude and longitude of the centre point of the site for each alternative site. The co-ordinates should be in degrees and decimal minutes. The minutes should have at least three decimals to ensure adequate accuracy. The projection that must be used in all cases is the WGS84 spheroid in a national or local projection.

List alternative sites, if applicable:

Alternative:

Alternative S1¹ (preferred or only site alternative)

Alternative S2 (if any)

Alternative S3 (if any)

Latitude (S):

Longitude (E):

33°	55'36.54"	25°	34'28.27"
⊖	⊖	⊖	⊖
⊖	⊖	⊖	⊖

In the case of linear activities:

Alternative:

Alternative S1 (preferred or only route alternative)

• Starting point of the activity

• Middle point of the activity

• End point of the activity

Alternative S2 (if any)

• Starting point of the activity

• Middle point of the activity

• End point of the activity

Alternative S3 (if any)

• Starting point of the activity

• Middle point of the activity

• End point of the activity

Latitude (S):

Longitude (E):

⊖	⊖	⊖	⊖
⊖	⊖	⊖	⊖
⊖	⊖	⊖	⊖

⊖	⊖	⊖	⊖
⊖	⊖	⊖	⊖
⊖	⊖	⊖	⊖

⊖	⊖	⊖	⊖
⊖	⊖	⊖	⊖
⊖	⊖	⊖	⊖

¹ “Alternative S..” refer to site alternatives.

For route alternatives that are longer than 500 m, please provide an addendum with co-ordinates taken every 250 m along the route for each alternative alignment.

4. PHYSICAL SIZE OF THE ACTIVITY

Indicate the physical size of the preferred activity/ technology as well as alternative activities/ technologies (footprints):

Alternative:

Alternative A1² (preferred activity alternative)

Alternative A2 (if any)

Alternative A3 (if any)

or, for linear activities:

Alternative:

Alternative A1 (preferred activity alternative)

Alternative A2 (if any)

Alternative A3 (if any)

Size of the activity:

350 m ²
m ²
m ²

Length of the activity:

m
m
m

Indicate the size of the alternative sites or servitudes (within which the above footprints will occur):

Alternative:

Alternative A1 (preferred activity alternative)

Alternative A2 (if any)

Alternative A3 (if any)

Size of the site/ servitude:

55 000 m ²
m ²
m ²

5. SITE ACCESS

Does ready access to the site exist?

If NO, what is the distance over which a new access road will be built

Describe the type of access road planned:

YES	
m	

Include the position of the access road on the site plan and required map, as well as an indication of the road in relation to the site.

6. SITE OR ROUTE PLAN

A detailed site or route plan(s) must be prepared for each alternative site or alternative activity. It must be attached as **Appendix A** to this document.

The site or route plans must indicate the following:

- 6.1 the scale of the plan which must be at least a scale of 1:500;
- 6.2 the property boundaries and numbers of all the properties within 50 m of the site;
- 6.3 the current land use as well as the land use zoning of each of the properties adjoining the site or sites;
- 6.4 the exact position of each element of the application as well as any other structures on the site;
- 6.5 the position of services, including electricity supply cables (indicate above or underground), water supply pipelines, boreholes, streetlights, sewage pipelines, stormwater infrastructure and telecommunication infrastructure;
- 6.6 all trees and shrubs taller than 1.8 m;
- 6.7 walls and fencing including details of the height and construction material;
- 6.8 servitudes indicating the purpose of the servitude;
- 6.9 sensitive environmental elements within 100 m of the site or sites including (but not limited thereto):
 - rivers;
 - the 1:100-year flood line (where available or where it is required by the Department of Water Affairs);

² "Alternative A.." refer to activity, process, technology or other alternatives.

- ridges;
 - cultural and historical features;
 - areas with indigenous vegetation (even if it is degraded or invested with alien species);
- 6.9 for gentle slopes the 1 m contour intervals must be indicated on the plan and whenever the slope of the site exceeds 1:10, the 500 mm contours must be indicated on the plan; and
- 6.10 the positions from where photographs of the site were taken.

7. SITE PHOTOGRAPHS

Colour photographs from the centre of the site must be taken in at least the eight major compass directions with a description of each photograph. Photographs must be attached under **Appendix B** to this form. It must be supplemented with additional photographs of relevant features on the site, if applicable.

8. FACILITY ILLUSTRATION

A detailed illustration of the activity must be provided at a scale of 1:200 as **Appendix C** for activities that include structures. The illustrations must be to scale and must represent a realistic image of the planned activity. The illustration must give a representative view of the activity.

9. ACTIVITY MOTIVATION

9(a) Socio-economic value of the activity

What is the expected capital value of the activity on completion?

What is the expected yearly income that will be generated by or as a result of the activity?

Will the activity contribute to service infrastructure?

Is the activity a public amenity?

~~How many new employment opportunities will be created in the development phase of the activity?~~

~~What is the expected value of the employment opportunities during the development phase?~~

~~What percentage of this will accrue to previously disadvantaged individuals?~~

How many permanent new employment opportunities will be created during the operational phase of the activity?

What is the expected current value of the employment opportunities during the first 10 years?

What percentage of this will accrue to previously disadvantaged individuals?

R 90 000 000.00	
R 21 000 000.00	
	NO
	NO
R	
%	
5	
R 17 500 000.00	
80%	

9(b) Need and desirability of the activity

Motivate and explain the need and desirability of the activity (including demand for the activity):

The need and desirability of lead-acid batteries in South Africa, in addition to use in industrial activities and motor vehicles, are driven by several factors:

- Renewable energy integration: The country is actively pursuing renewable energy integration to diversify its energy mix and reduce reliance on fossil fuels. Lead-acid batteries offer a cost-effective solution for storing energy generated from renewable sources, ensuring grid stability and reliability.
- Electric mobility transition: The automotive industry is transitioning towards electric/ hybrid mobility, with advanced lead-acid batteries playing a significant role in hybrid vehicles, providing a proven and economical energy storage solution.
- Telecommunications and rural electrification: The expanding telecommunications sector and rural electrification initiatives create additional demand for reliable energy storage solutions, where advanced lead-acid batteries can support off-grid power systems and backup power solutions.
- Sustainability and recycling: The recyclability of lead-acid batteries aligns with South Africa's commitment to

sustainability and environmental conservation, contributing to waste reduction efforts and supporting the circular economy.

Overall, the advanced lead-acid battery market in South Africa is positioned as a sector with significant potential for growth and development, driven by renewable energy goals, electric mobility trends, and sustainability objectives.

South Africa is heavily reliant on exported products which has a potential to undermine the local production of materials. Auto-X plays a crucial role in keeping production within the domestic economy. This has a direct impact on local job and skills-development and retention. This does not only apply to the facility workers, but mining industry where the Pb alloy is sourced and truck drivers that evacuate the completed batteries. This does not include the many other stakeholders that are involved in this industry.

There is no doubt that the Eastern Cape Province needs economic opportunities and skills enhancement. When properly handled, which Auto-X has been doing for several years, lead-acid battery assembly displays more benefits than negatives.

South Africa does not only cater to the domestic market with regards to batteries, but the international market as well.

Indicate any benefits that the activity will have for society in general:

The proposed development will accrue the following benefits:

- Retainment of current job and skills development including transfer;
- Employment of five additional personnel; and
- High-quality lead-acid batteries to the local and international market.

Other benefits comprise:

- Cost-effectiveness: Lead-acid batteries are relatively inexpensive to manufacture and maintain, making them a cost-effective solution for many applications.
- Reliability: They are reliable and can function in a wide range of temperatures and conditions, ensuring consistent power supply.
- Recyclability: They have a high recycling rate, reducing the environmental impact, and promoting sustainable resource use.
- Versatility: Lead-acid batteries are compatible with current infrastructure, allowing seamless integration into systems without the need for costly modifications.

These advantages make lead-acid batteries a vital component in supporting national security, transportation, communications, and climate mitigation efforts. They play a critical role in powering everyday life and essential infrastructure, ensuring that many systems people rely on daily function effectively.

Indicate any benefits that the activity will have for the local communities where the activity will be located:

The local community is currently employed within the facility, and five more jobs are envisaged as part of the BM Line 3 operation. Thus, the benefits will be through economic development, employment retainment and generation, skills development and transfer, and poverty eradication to the affected personnel.

10. APPLICABLE LEGISLATION, POLICIES AND/OR GUIDELINES

List all legislation, policies and/or guidelines of any sphere of government that are applicable to the application as contemplated in the Regulations, if applicable:

Title of legislation, policy or guideline:	Administering authority:	Date:
National Environmental Management Act 107 of 1998, as amended	The Department	1999
Environmental Impact Assessment Regulations of 2014, as amended	The Department	2014
National Environmental Management Air Quality Act 39 of 2004, as amended	NMBM	2005
Listed Activities and associated Minimum Emission Standards of 2013, as amended	NMBM	2013
National Environmental Management Waste Act 59 of 2008 as amended (NEMWA)	The Department	2009
Occupational Health and Safety Act 85 of 1993, as amended	Department of Labour	1994
National Health Act 61 of 2003, as amended	Department of Health	2005
SANS 1929:2005 (Ambient Air Quality Management)	NMBM	2005
NMBM: Air Pollution Control By-Law	NMBM	2010
NMBM Integrated Development Plan	NMBM	2022/27
NMBM By-Laws (Waste Management, Solid Waste, Water and Sanitation)	NMBM	Various

11. WASTE, EFFLUENT, EMISSION AND NOISE MANAGEMENT

11(a) Solid waste management

Will the activity produce solid construction waste during the construction/ initiation phase? ☒ YES ☐ NO

If yes, what estimated quantity will be produced per month?

How will the construction solid waste be disposed of (describe)?

Where will the construction solid waste be disposed of (describe)?

Will the activity produce solid waste during its operational phase? ☒ YES ☐ NO

If yes, what estimated quantity will be produced per month?

How will the solid waste be disposed of (describe)?

All waste generated on site as per the existing operations is handled by the NMBM (general) and licenced, private service providers (hazardous). The waste to be generated by BM Line 3 will be managed through these existing and approved waste management systems. This includes packaging waste (cardboard, plastic, wood pallets) from incoming machinery and component materials. It also has a robust recycling system for cardboard, plastic, lead and steel.

Where will the solid waste be disposed of if it does not feed into a municipal waste stream (describe)?

As above, private service providers.

If the solid waste (construction or operational phases) will not be disposed of in a registered landfill site or be taken up in a municipal waste stream, then the applicant should consult with the CA to determine whether it is necessary to change to an application for S&EIA.

Can any part of the solid waste be classified as hazardous in terms of the relevant legislation? ☒ YES ☐ NO

If yes, inform the CA and request a change to an application for S&EIA.

The EA application pertains to the amendment of existing operations. The hazardous waste generated by BM Line 3 will be limited to spent filter cartridges from the new Donaldson bag filter. Estimated generation is negligible, approximately two to four cartridges per annum. These will be managed through Auto-X's existing licenced hazardous waste service provider. This does not constitute a S&EIA in terms of the NEMWA.

Is the activity that is being applied for a solid waste handling or treatment facility? ☒ YES ☐ NO

If yes, then the applicant should consult with the CA to determine whether it is necessary to change to an application for S&EIA.

11(b) Liquid effluent

Will the activity produce effluent, other than normal sewage, that will be disposed of in a municipal sewage system? ☒ YES ☐ NO

If yes, what estimated quantity will be produced per month?

Will the activity produce any effluent that will be treated and/or disposed of on-site? ☒ YES ☐ NO

If yes, the applicant should consult with the CA to determine whether it is necessary to change to an application for S&EIA.

Will the activity produce effluent that will be treated and/or disposed of at another facility? ☒ YES ☐ NO

If yes, provide the particulars of the facility:

Facility name:	NMBM wastewater infrastructure		
Contact person:	Teresa Wiegand		
Postal address:	P.O Box 11, Central, Port Elizabeth		
Postal code:	6000		
Telephone:	041 505 4578	Cell:	066 241 9322
E-mail:	twiegand@mandelametro.gov.za	Fax:	

Describe the measures that will be taken to ensure the optimal reuse or recycling of wastewater, if any:

BM Line 3 will generate process effluent from the battery assembly and washing operations. This effluent will be discharged to the municipal sewerage system in accordance with Auto-X's existing Trade Effluent Discharge Permit issued by the NMBM. The anticipated additional volume, estimated at 10-15 m³ per month, is within the permitted capacity of the existing permit.

Auto-X operates a reverse osmosis plant to improve the quality of water for application within the site. This plant also assists in recycling of water within some areas of the operations.

11(c) Emissions into the atmosphere

Will the activity release emissions into the atmosphere?

YES

If yes, is it controlled by any legislation of any sphere of government?

YES

If yes, the applicant should consult with the CA to determine whether it is necessary to change to an application for S&EIA.

The Applicant is currently a holder of an AEL issued by the NMBM, a variation application in this regard (to take BM Line 3 into account) has been submitted to the NMBM.

If no, describe the emissions in terms of type and concentration:

11(d) Generation of noise

Will the activity generate noise?

YES

If yes, is it controlled by any legislation of any sphere of government?

NO

If yes, the applicant should consult with the CA to determine whether it is necessary to change to an application for S&EIA.

If no, describe the noise in terms of type and level:

Noise in battery assembly lines can be caused by various factors, including the operation of automated assembly systems, robotic welding machines, conveyor lines, and inspection stations. These machines, when operating simultaneously, can lead to high sound levels.

Mitigation measures, i.e. noise absorbers, are in place throughout the facility to absorb sound energy, improving overall comfort in the battery assembly environment. Additionally, it was noted during the site investigation, that the workers around noise generating areas are equipped with noise ear blockers. It is the understanding of the Reg. EAP that this practice will continue on site.

12. WATER USE

Please indicate the source(s) of water that will be used for the activity by ticking the appropriate box(es)

municipal	groundwater	other
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If water is to be extracted from groundwater, river, stream, dam, lake or any other natural feature, please indicate the volume that will be extracted per month:

1.2M litres per year
(groundwater + municipal +
other: rainwater)

Does the activity require a water use permit from the Department of Water Affairs?

NO

This volume is within the existing municipal water account capacity and issued WUL. Auto-X currently harvests rainwater for non-potable uses, the proposed line will utilise the existing rainwater harvesting system. No additional abstraction point is planned. As such, no new water use authorisation is required nor existing WUL amendment.

If yes, please submit the necessary application to the Department of Water Affairs and attach proof thereof to this

application if it has been submitted.

13. ENERGY EFFICIENCY

Describe the design measures, if any, that have been taken to ensure that the activity is energy efficient:

Various measures are implemented within the existing facility as part of saving energy; these include insulated chilled water pipelines to keep the water cool such that less energy is required for cooling; use of energy efficient generators; light-emitting diodes/ LED; traditional geysers were replaced with solar heaters; etc. As such, Auto-X is a holder of an ISO 50001 Certificate issued in 2023 for energy management.

Describe how alternative energy sources have been taken into account or been built into the design of the activity, if any:

The operations currently utilise LPG for the heavily consumptive activities on site, not only to lessen the strain on the national grid but to save on costs as well. This practice is planned to continue for the foreseeable future.

SECTION B: SITE/AREA/PROPERTY DESCRIPTION

Important notes:

1. For linear activities (pipelines, etc.) as well as activities that cover very large sites, it may be necessary to complete this section for each part of the site that has a significantly different environment. In such cases please complete copies of Section C and indicate the area, which is covered by each copy No. on the Site Plan.
Section C Copy No. (e.g. A):
2. Paragraphs 1–6 below must be completed for each alternative.
3. Has a specialist been consulted to assist with the completion of this section?

NO
- If YES, please complete form XX for each specialist thus appointed:
All specialist reports must be contained in Appendix D.

1. GRADIENT OF THE SITE

Indicate the general gradient of the site.

Alternative S1:

Flat

Alternative S2 (if any):

Flat	1:50–1:20	1:20–1:15	1:15–1:10	1:10–1:7,5	1:7,5–1:5	Steeper than 1:5
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Alternative S3 (if any):

Flat	1:50–1:20	1:20–1:15	1:15–1:10	1:10–1:7,5	1:7,5–1:5	Steeper than 1:5
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2. LOCATION IN LANDSCAPE

Indicate the landform(s) that best describes the site:

2.6 Plain

3. GROUNDWATER, SOIL, AND GEOLOGICAL STABILITY OF THE SITE

Is the site(s) located on any of the following (tick the appropriate boxes)?

	Alternative S1:	Alternative S2 (if any):	Alternative S3 (if any):
Shallow water table (less than 1.5 m deep)	YES	YES NO	YES NO
Dolomite, sinkhole or doline areas	NO	YES NO	YES NO
Seasonally wet soils (often close to water bodies)	NO	YES NO	YES NO
Unstable rocky slopes or steep slopes with loose soil	NO	YES NO	YES NO
Dispersive soils (soils that dissolve in water)	NO	YES NO	YES NO
Soils with high clay content (clay fraction more than 40%)	YES	YES NO	YES NO
Any other unstable soil or geological feature	NO	YES NO	YES NO
An area sensitive to erosion	NO	YES NO	YES NO

If you are unsure about any of the above or if you are concerned that any of the above aspects may be an issue of concern in the application, an appropriate specialist should be appointed to assist in the completion of this section. (Information in respect of the above will often be available as part of the project information or at the planning sections of local authorities. Where it exists, the 1:50 000 scale Regional Geotechnical Maps prepared by the Council for Geo Science may also be consulted).

4. GROUNDCOVER

Indicate the types of groundcover present on the site:

4.8 Paved surface

4.9 Building or other structure

The location of all identified rare or endangered species or other elements should be accurately indicated on the site plan(s).

If any of the boxes marked with an "E" is ticked, please consult an appropriate specialist to assist in the completion of this section if the EAP doesn't have the necessary expertise.

5. LAND USE CHARACTER OF SURROUNDING AREA

Indicate land uses and/or prominent features that currently occur within a 500 m radius of the site and give description of

how this influences the application or may be impacted upon by the application:

5.3 Medium density residential	
5.4 High density residential	
5.6 Retail commercial & warehousing	
5.7 Light industrial	
5.8 Medium industrial ^{AN}	
5.9 Heavy industrial ^{AN}	
5.11 Office/consulting room	
5.16 Hospital/medical centre	
5.19 Church	
5.23 Railway line ^N	
5.27 Sport facilities	
5.30 Filling station ^H	
5.34 River, stream or wetland	

If any of the boxes marked with an "N "are ticked, how will this impact/ be impacted upon by the proposed activity.

A defunct railway line exists to the north and east of the existing operations, there was no indication that any effect, vice versa, exists.
--

If any of the boxes marked with an "An" are ticked, how will this impact / be impacted upon by the proposed activity.

If YES, specify and explain:	The operations exist amongst other industrial activities, and there was no indication that any effect on one another exists.
If YES, specify:	

If any of the boxes marked with an "H" are ticked, how will this impact / be impacted upon by the proposed activity.

If YES, specify and explain:	A filling station is located to the south of the existing operations, during the site investigation undertaken on 24 June 2026 there was no indication that there were effects on each other as a result of both operations.
If YES, specify:	

6. CULTURAL/ HISTORICAL FEATURES

Are there any signs of culturally or historically significant elements, as defined in section 2 of the National Heritage Resources Act 25 of 1999, including Archaeological or palaeontological sites, on or close (within 20 m) to the site?

NO

If yes, explain:

If uncertain, conduct a specialist investigation by a recognised specialist in the field to establish whether there is such a feature(s) present on or close to the site.

Briefly explain the findings of the specialist:

Will any building or structure older than 60 years be affected in any way?

YES

NO

Is it necessary to apply for a permit in terms of the National Heritage Resources Act 25 of 1999?

YES

NO

If yes, please submit or, make sure that the applicant or a specialist submits the necessary application to SAHRA or the relevant provincial heritage agency and attach proof thereof to this application if such application has been made.

SECTION C: PUBLIC PARTICIPATION

1. ADVERTISEMENT

The person conducting a public participation process (PPP) must take into account any guidelines applicable to public participation as contemplated in section 24J of the Act and must give notice to all potential I&APs of the application which is subjected to public participation by:

- fixing a notice board (of a size at least 60 centimetres [cm] by 42 cm; and must display the required information in lettering and in a format as may be determined by the CA) at a place conspicuous to the public at the boundary or on the fence of:
 - the site where the activity to which the application relates is or is to be undertaken; and
 - any alternative site mentioned in the application;
- giving written notice to:
 - the owner or person in control of that land if the applicant is not the owner or person in control of the land;
 - the occupiers of the site where the activity is or is to be undertaken or to any alternative site where the activity is to be undertaken;
 - owners and occupiers of land adjacent to the site where the activity is or is to be undertaken or to any alternative site where the activity is to be undertaken;
 - the municipal councillor of the ward in which the site or alternative site is situated and any organisation of ratepayers that represent the community in the area;
 - the municipality which has jurisdiction in the area;
 - any organ of state having jurisdiction in respect of any aspect of the activity; and
 - any other party as required by the CA;
- placing an advertisement in:
 - one local newspaper; or
 - any official *Gazette* that is published specifically for the purpose of providing public notice of applications or other submissions made in terms of the Regulations;
- placing an advertisement in at least one provincial newspaper or national newspaper, if the activity has or may have an impact that extends beyond the boundaries of the metropolitan or local municipality in which it is or will be

- undertaken: Provided that this paragraph need not be complied with if an advertisement has been placed in an official *Gazette* referred to in subregulation 54(c)(ii); and
- (e) using reasonable alternative methods, as agreed to by the CA, in those instances where a person is desiring of but unable to participate in the process due to:
- (i) illiteracy;
 - (ii) disability; or
 - (iii) any other disadvantage.

2. CONTENT OF ADVERTISEMENTS AND NOTICES

A notice board, advertisement or notices must:

- (a) indicate the details of the application which is subjected to public participation; and
- (b) state:
 - (i) that the application has been submitted to the CA in terms of the Regulations, as the case may be;
 - (ii) whether basic assessment or scoping procedures are being applied to the application, in the case of an application for EA;
 - (iii) the nature and location of the activity to which the application relates;
 - (iv) where further information on the application or activity can be obtained; and
 - (v) the manner in which and the person to whom representations in respect of the application may be made.

3. PLACEMENT OF ADVERTISEMENTS AND NOTICES

Where the proposed activity may have impacts that extend beyond the municipal area where it is located, a notice must be placed in at least one provincial newspaper or national newspaper, indicating that an application will be submitted to the CA in terms of the Regulations, the nature and location of the activity, where further information on the proposed activity can be obtained and the manner in which representations in respect of the application can be made, unless a notice has been placed in any *Gazette* that is published specifically for the purpose of providing notice to the public of applications made in terms of the Regulations.

Advertisements and notices must make provision for all alternatives.

4. DETERMINATION OF APPROPRIATE MEASURES

The practitioner must ensure that the public participation is adequate and must determine whether a public meeting or any other additional measure is appropriate or not based on the particular nature of each case. Special attention should be given to the involvement of local community structures such as Ward Committees, ratepayers associations and traditional authorities where appropriate. Please note that public concerns that emerge at a later stage that should have been addressed may cause the competent authority to withdraw any authorisation it may have issued if it becomes apparent that the public participation process was inadequate.

5. COMMENTS AND RESPONSE REPORT

The practitioner must record all comments and respond to each comment of the public before the application is submitted. The comments and responses must be captured in a comments and response report as prescribed in the Regulations and be attached to this application. The comments and response report (CRR) must be attached under **Appendix E**.

6. AUTHORITY PARTICIPATION

Authorities are key I&APs in each application and no decision on any application will be made before the relevant local authority is provided with the opportunity to give input. The planning and the environmental sections of the local authority must be informed of the application at least 30 (thirty) calendar days before the submission of the application.

List of authorities informed:

This is an ongoing process, final list to be included following the PPP.

List of authorities from whom comments have been received:

This is an ongoing process, final list to be included following the PPP.

7. CONSULTATION WITH OTHER STAKEHOLDERS

Note that, for linear activities, or where deviation from the public participation requirements may be appropriate, the person conducting the PPP may deviate from the requirements of that subregulation to the extent and in the manner as may be agreed to by the CA.

Any stakeholder that has a direct interest in the site or property, such as servitude holders and service providers, should be informed of the application at least 30 (thirty) calendar days before the submission of the application and be provided with the opportunity to comment.

Has any comment been received from stakeholders?

YES NO

If "YES", briefly describe the feedback below (also attach copies of any correspondence to and from the stakeholders):

This is an ongoing process; a brief description of the feedback is to be included following the PPP.

SECTION D: IMPACT ASSESSMENT

The assessment of impacts must adhere to the minimum requirements in the Regulations and should take applicable official guidelines into account. The issues raised by I&APs should also be addressed in the assessment of impacts.

1. ISSUES RAISED BY INTERESTED AND AFFECTED PARTIES

List the main issues raised by I&APs:

The draft report is currently out for PPP, I&APs are yet to provide comments.

Response from the practitioner to the issues raised by the I&APs (a full response must be given in the CRR that must be attached to this report):

Comments are yet to be provided; the EAP will attend to such as soon as provided and also respond in the CRR.

2. IMPACTS THAT MAY RESULT FROM THE PLANNING AND DESIGN, ~~CONSTRUCTION,~~ OPERATIONAL, DECOMMISSIONING AND CLOSURE PHASES AS WELL AS PROPOSED MANAGEMENT OF IDENTIFIED IMPACTS, AND PROPOSED MITIGATION MEASURES

List the potential direct, indirect, and cumulative property/ activity/ design/ technology/ operational alternative related impacts (as appropriate) that are likely to occur as a result of the planning and design phase, ~~construction phase,~~ operational phase, decommissioning and closure phase (although not planned by the applicant), including impacts relating to the choice of site/ activity/ technology alternatives as well as the mitigation measures that may eliminate or reduce the potential impacts listed.

2.1. Potential Impacts

Alternative (preferred alternative)

2.1.1. Planning and Design Phase

Direct impacts:

- Ensure any new job opportunities accrue to the local community.

- Make all necessary arrangements with applicable stakeholders to ascertain that all new line requirements are met.

Indirect impacts:

- Ensure the battery assembly machinery technology is energy efficient than the existing.

Cumulative impacts:

- None anticipated.

2.1.2. Operational Phase

Direct impacts:

- Release of Pb and H₂SO₄ into the environment.
- Release of fumes/ air emissions due to the welding/ burning associated with the battery assembly.
- Energy consumption.
- GHG emissions.
- Job retention.

Indirect impacts:

- Environmental pollution, including air pollution, and health risks associated with Pb, H₂SO₄, and fumes.
- Localised, around some machinery, elevated noise levels.
- Skills development and transfer.
- Employment creation and skills development associated with battery recycling and other second-life applications.

Cumulative impacts:

- Release of Pb and H₂SO₄ into the environment.
- Production of H₂SO₄ can cause acid rain which has a negative effect to soil, plants, and aquatic life.
- The mining of Pb can result in environmental degradation.
- Depletion of finite resources, i.e. Pb alloy.

2.1.3. Decommissioning and Closure Phase

Direct impacts:

- Free up of energy for other uses.
- Job losses.
- Negative effect in the lead-acid battery market.

Indirect impacts:

- Lessened environmental pollution, including air pollution.
- Skills and skills transfer loss.

Cumulative impacts:

- Job losses.

2.1.4. No-Go

- Status-quo; no BM Line 3, but continuation of BM Lines 1 and 2.
- No additional job creation, skills development and transfer, and economic inflow.
- The other associated impacts will still continue as per the existing operations, save for the economic impacts pertaining to additional jobs.

2.2. Impact Assessment Methodology

Abantu has utilised its own Impact Assessment Methodology detailed in this section to assess the potential impacts

identified as per this section.

The impacts identified were assessed according to the criteria outlined in the tables below. Each impact was ranked according to the nature, extent, duration, magnitude, probability, irreplaceable loss of resources and reversibility. These criteria are based on the DEDEAT Guideline Document to the EIA Regulations (1998). A significance rating of each identified impact was further reviewed by the *Reg. EAP* and/or air quality specialist by applying professional judgement. Mitigatory measures were recommended for the impacts identified.

Rating	Definition of Rating	Score
A. Spatial extent – the area in which the impact will be experienced		
None		0
Local	Confined to project or study area or part thereof (e.g. site)	1
Regional	The region, which may be defined in various ways, e.g. cadastral, catchment, topographic	2
(Inter)national	Nationally or beyond	3
B. Intensity – the magnitude or size of the impact		
None		0
Low	Natural and/or social functions and processes continue albeit in a slightly modified way	1
Medium	Natural and/or social functions and processes continue albeit in a modified way	2
High	Natural and/or social functions and processes are severely altered	3
C. Duration – the time frame for which the impact will be experienced		
None		0
Short-term	Up to 2 years	1
Medium-term	2 to 20 years	2
Long-term	More than 20 years	3

Combined Score (A+B+C)	
Consequence Rating	
Not significant	0 - 2
Very low	3 - 4
Low	5
Medium	6
High	7
Very high	8 - 9

Probability – the likelihood of the impact occurring		
Improbable	< 40% chance of occurring	0
Possible	40% - 70% chance of occurring	1
Probable	> 70% - 90% chance of occurring	2
Definite	> 90% chance of occurring	3

Significance Rating	Possible Impact Combinations	
	Consequence	Probability
Insignificant	Very Low	Improbable
	Very Low	Possible
Very Low	Very Low	Probable
	Very Low	Definite
	Low	Improbable
	Low	Possible
Low	Low	Probable
	Low	Definite

Significance Rating	Possible Impact Combinations	
	Consequence	Probability
	Medium	Improbable
	Medium	Possible
Medium	Medium	Probable
	Medium	Definite
	High	Improbable
	High	Possible
High	High	Probable
	High	Definite
	Very High	Improbable
	Very High	Possible
Very High	Very High	Probable
	Very high	Definite

Status of impact	
<i>The environmental impacts of an activity are those resultant changes in environmental parameters, in space and time, compared with what would have happened had the activity not been undertaken. It is an appraisal of the type of effect the activity would have on the affected environmental parameter. Its description includes what is being affected, and how</i>	
Indication whether the impact is adverse (negative) or beneficial (positive).	+ ve (positive – a 'benefit')
	– ve (negative – a 'cost')

Confidence of assessment	
The degree of confidence in predictions based on available information, EAP's and/or specialist knowledge	Low
	Medium
	High

2.3. Impact Assessment

All potential impacts associated with the preferred alternative have been categorised according to the respective development phases during which they will occur, planning and design, operation, decommissioning and closure (although not planned by the applicant). Potential impacts identified as being associated with the preferred alternative have been outlined and assessed in terms of their anticipated duration, extent, severity, probability and significance (**Table 4** below), both prior and post mitigation measures being implemented, making use of the methodology described in this section, above.

2.3.1. Preferred Alternative

Table 4: Impact assessment

Impact	Mitigation	Extent	Intensity	Duration	Consequence	Consequence	Probability	Probability	Significance	Status	Confidence
Direct											
Job opportunities	Without	2	1	2	5	Low	3	Definite	Low	+ ve	Medium
	With	2	2	2	6	Medium	3	Definite	Medium	+ ve	Medium
BM Line 3 permitting requirements	Without	2	2	2	6	Medium	1	Possible	Low	- ve	High
	With	1	1	1	3	Very low	0	Improbable	Insignificant	- ve	High
Indirect											
Source energy efficient machinery	Without	2	1	2	5	Low	2	Probable	Low	+ ve	High
	With	2	2	2	6	Medium	3	Definite	Medium	+ ve	High
Direct											
Release of Pb and H ₂ SO ₄ into the environment	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	High
	With	1	1	2	4	Very low	2	Probable	Very Low	- ve	High
Release of fumes/ air emissions due the welding/ burning associated with BM Line 3	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	High
	With	1	1	2	4	Very low	2	Probable	Very Low	- ve	High
Energy consumption	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	High
	With	1	1	2	4	Very low	2	Probable	Very Low	- ve	High
GHG emissions	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	High
	With	1	1	2	4	Very low	2	Probable	Very Low	- ve	High
Job retention	Without	1	2	1	4	Very low	2	Probable	Very Low	+ ve	High
	With	2	2	2	6	Medium	3	Definite	Medium	+ ve	High
Indirect											
Environmental pollution, including air pollution, and health risks associated with Pb, H ₂ SO ₄ , and fumes	Without	2	2	2	6	Medium	2	Probable	Medium	- ve	High
	With	1	1	2	4	Very low	1	Possible	Insignificant	-ve	High
Localised, around some machinery, elevated noise levels	Without	1	1	3	5	Low	2	Probable	Low	- ve	High
	With	1	1	2	4	Very low	1	Possible	Insignificant	- ve	High
Skills development and transfer	Without	1	1	3	5	Low	2	Probable	Low	+ ve	High
	With	2	2	2	6	Medium	3	Definite	Medium	+ ve	High
Employment and skills development associated with battery recycling and other second-life applications	Without	1	1	3	5	Low	2	Probable	Low	+ ve	High
	With	2	2	2	6	Medium	3	Definite	Medium	+ ve	High
Cumulative											
Release of Pb and H ₂ SO ₄ into the environment	Without	2	2	2	6	Medium	2	Probable	Medium	- ve	High
	With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
Production of H ₂ SO ₄ can cause acid rain which has a negative effect to soil, plants, and aquatic life	Without	2	2	2	6	Medium	2	Probable	Medium	- ve	High
	With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
The mining of Pb can result in environmental degradation	Without	2	2	2	6	Medium	2	Probable	Medium	- ve	High
	With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
Depletion of finite resources, i.e. Pb alloy	Without	2	2	2	6	Medium	2	Probable	Medium	- ve	High
	With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
Direct											

Impact	Mitigation	Extent	Intensity	Duration	Consequence	Consequence	Probability	Probability	Significance	Status	Confidence
Free up of energy for other uses	Without	1	1	1	3	Very low	1	Possible	Insignificant	+ ve	Medium
	With	2	2	2	6	Medium	2	Probable	Medium	+ ve	Medium
Job losses	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	High
	With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
Negative effect on the lead-acid battery market	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	Medium
	With	1	1	1	3	Very low	2	Probable	Very Low	- ve	Medium
Direct											
Lessened environmental pollution, including air pollution	Without	1	1	2	4	Very low	1	Possible	Insignificant	+ ve	Medium
	With	2	2	3	7	High	2	Probable	High	+ ve	Medium
Skills and skills transfer loss	Without	2	2	2	6	Medium	2	Probable	Medium	- ve	High
	With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
Indirect											
Job losses	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	High
	With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
Direct											
Status-quo; no BM Line 3 addition, but continuation of BM Lines 1 and 2	Without	2	2	2	6	Medium	2	Probable	Medium	- ve	High
	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	High
No additional job creation, skills development and transfer, and economic inflow	Without	2	2	2	6	Medium	2	Probable	Medium	- ve	High
	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	High
The other associated impacts will still continue as per the existing operations, save for the economic impacts	Without	2	2	2	6	Medium	2	Probable	Medium	- ve	High
	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	High

2.4. Management and Mitigation Measures

By adopting these measures, the socio-economic and environmental impact of lead-acid battery assembly can be significantly reduced, contributing to a more sustainable energy storage solution.

Alternative (preferred alternative)

2.4.1. Planning and Design Phase

- Ensure any new job opportunities accrue to the local community.
- Make all necessary arrangements with applicable stakeholders to ascertain that all new line requirements are met.
- Ensure the battery assembly machinery technology is energy efficient than the existing machinery.
- By prioritising energy efficiency in battery assembly, manufacturers can achieve both economic and environmental benefits, contributing to a more sustainable energy future.
- Energy-efficient processes minimise the amount of energy used during the production of batteries, leading to lower operational costs.

2.4.2. Operational Phase

- Minimise the emissions of Pb, H₂SO₄, and any other materials utilised within the facility.
- Regular testing/ sampling of Pb levels to ascertain emissions are minimised. Periodic ambient Pb monitoring must be considered in the easterly corner of the facility or neighbouring property to verify Pb emissions against the NAAQS.
- Strengthen policies and plans to prevent Pb exposure, including the adoption of standards and strong implementation of these measures.
- Duct audit should be undertaken frequently.
- Continue to educate and provide refresher talks on safely handling of the materials used on the facility such that that do not become a danger to the workers.
- Follow the World Health Organisation (WHO) developments on any initiatives and further measures for safely working with Pb.
- Immediate action following any accidental release of H₂SO₄ involves containment to prevent further spread and neutralisation to mitigate corrosive effects, often employing alkaline agents such as lime or soda ash.
- Long-term action necessitates the implementation of improved safety protocols, including enhanced storage facility design, employee training, and emergency response drills.
- Thorough analysis of H₂SO₄ release incidents is essential for identifying root causes, evaluating the effectiveness of response measures, and informing preventative strategies.
- Continue to apply the energy saving measures discussed above, i.e. insulated chilled water pipelines to keep the water cool such that less energy is required for cooling; use of energy efficient generators; light-emitting diodes/ LED; supplementing the NMBM provided electricity with LPG, etc.
- Optimise industrial processes, where feasible, and implementing smart technologies for demand management.
- Conduct comprehensive energy audits to identify high-consumption areas and implement targeted efficiency upgrades.
- Encourage management and workers behaviour changes and appliance upgrades to minimise demand.
- Effective mitigation results in verifiable reductions in operational costs and associated GHGs, improving corporate sustainability metrics.
- Continue to conduct comprehensive lifecycle assessments to quantify and mitigate GHG footprints associated with battery production, including material sourcing and manufacturing.
- Continue with the extended producer responsibility practices currently implemented.
- Establish clear supply chain transparency and ensuring adherence to environmental and labour standards throughout the entire value chain.
- Foster skilled jobs in battery recycling and second-life applications, which can positively impact local economies.
- Continue to enhance local manufacturing capacity, job creation and retention, while also supporting the energy

value chain.

- Consider closed-loop recycling, where feasible. This involves processing used lead-acid batteries to extract Pb and H₂SO₄, using these materials to manufacture new batteries. It lessens the impact on the environment and the need for additional raw resources.
- Ongoing research focuses on developing advanced recycling technologies that are more energy-efficient and environmentally friendly. Innovations in hydrometallurgical and pyrometallurgical processes aim to enhance recycling efficiency and minimise emissions.
- Promote responsible disposal practices to minimise environmental harm.
- Continue to adhere to policies and regulations governing the proper disposal and recycling of lead-acid batteries.
- Implement risk mitigation strategies, such as proper storage, transportation, and treatment of spent batteries, along with public education and awareness campaigns. This helps minimise environmental risks and ensure responsible battery management.
- Adhere to the Environmental Management Programme (EMPr) and ensure that contractors are trained in environmentally responsible practices.
- Continue with the monitoring of air quality, correct PPE use, and implementing air pollution abatement measures.
- Continue harvesting rainwater and implementing water saving measures.
- Continue implementing energy saving measurements.
- Continue to continuously engage with employees on the safety and health issues, i.e. the current use imagery around the facility and the toolbox talks.
- Continue with the monitoring of GHG emissions.
- Continue with the developed implementation of scheduled preventative maintenance programme.
- Conduct regular inspections using emissions leak detection technology.
- Ensure integration of the new assembly line into the existing battery manufacturing and assembly system.
- Establish a long-term maintenance and monitoring plan, including budget allocation.
- Develop robust recycling infrastructure and promote the circular economy to manage battery waste effectively.
- Ensure regular servicing, inspection, and upgrading of machinery for efficiency.
- Continue with the training of operators on rapid incident response and reporting protocols.

2.4.3. Decommissioning and Closure Phase

- Isolate all sources of electrical and mechanical energy to ensure safety.
- Physically disconnecting battery modules and other components.
- Package batteries in United Nations-approved, non-conductive containers with adequate padding and following hazardous material documentation.
- Direct recycling to recover valuable materials or repurposing for less demanding applications.
- Adhering to local, regional, and national regulations for waste handling and disposal.
- A formal decommissioning process is crucial for ensuring safety, regulatory compliance, and environmental responsibility. It standardises procedures across multiple sites, minimises liability risks, maximises resource recovery through recycling, and provides clear accountability at each stage from disconnection to final disposition or second-life use.
- Conduct a detailed site inspection and environmental due-diligence assessment before dismantling. Identify areas where Pb compounds, PbO, Pb paste, H₂SO₄, electrolyte, contaminated sludge, batteries, chemicals and wastes were handled. Review historical spills, waste-disposal areas, drainage systems and underground infrastructure. Establish baseline soil, groundwater, surface-water and, where relevant, indoor/ outdoor dust conditions.
- Survey floors, walls, roofs, machinery, ventilation systems, drains and process equipment for Pb contamination. Establish controlled work zones. Use methods that minimise dust generation during dismantling. Avoid dry sweeping and uncontrolled compressed-air cleaning; use wet cleaning or suitable HEPA-filtered industrial vacuum systems.
- Inventory all finished, defective, partially assembled and waste batteries. Segregate batteries from other wastes

and store them in a secure, covered and appropriately contained area. Prevent physical damage, short-circuiting and leakage. Send batteries to appropriately authorised recycling/ recovery facilities rather than ordinary landfill disposal. Lead-acid batteries should generally be recycled because improper disposal can release both Pb and H_2SO_4 .

- Drain, collect and manage residual electrolyte only using controlled procedures and suitable acid-resistant equipment. Do not discharge electrolyte to soil, stormwater drains or ordinary wastewater systems. Store recovered acid in compatible, labelled and bunded containers. Neutralisation or recovery should be undertaken by competent personnel using an approved procedure.
- Implement dust suppression during dismantling, demolition and material handling. Use wet methods where appropriate, local exhaust ventilation and HEPA filtration. Enclose particularly contaminated work areas. Keep potentially contaminated surfaces damp during disturbance and prevent windblown dust. WHO identifies Pb dust and particles as important exposure pathways during battery handling and recycling.
- Decontaminate machinery, conveyors, mixers, casting equipment, charging equipment, ventilation ducts, floors and other process equipment before removal from controlled areas. Characterise residues generated during cleaning and manage them according to their hazardous-waste classification.
- Inspect sumps, drains, interceptors and wastewater-treatment systems for Pb-bearing sludge and acidic residues. Collect contaminated liquids and sludges for appropriate treatment/ disposal. Prevent contaminated decommissioning water from entering stormwater systems.
- Develop a waste inventory and classify wastes before removal. Likely waste streams include Pb-bearing dust, sludge, contaminated PPE, spent filters, contaminated soil, electrolyte, chemicals, batteries, contaminated scrap and wastewater-treatment residues. Store wastes in labelled, compatible, sealed/ covered containers with secondary containment. Maintain waste manifests and disposal/ recycling records.
- Segregate clean scrap from Pb-contaminated scrap. Do not release contaminated metal as ordinary scrap until it has been appropriately characterised and, where required, decontaminated. Use authorised recyclers for Pb-containing materials.
- Investigate areas around battery storage, manufacturing lines, paste preparation, casting, charging, waste storage, loading areas, drainage channels and historical spill locations. If Pb contamination is identified, delineate the contamination spatially and vertically. Depending on risk and applicable regulatory criteria, options can include excavation and off-site treatment/ disposal, stabilisation/ solidification, containment, or other approved remediation technologies. The Environmental Protection Agency (United States) experience with lead-acid battery sites demonstrates the importance of assessing and remediating contaminated soil and groundwater.
- Investigate potential groundwater contamination where historical releases or permeable soils indicate a pathway. Prevent further infiltration from contaminated areas and establish groundwater monitoring where warranted by the site assessment.
- Separate clean and potentially contaminated stormwater during decommissioning. Cover stockpiles and waste-storage areas, maintain bunds and drainage controls, and inspect drainage after rainfall. Prevent sediment and Pb-contaminated runoff from leaving the site.
- Establish a Pb/ dust monitoring programme appropriate to the level of contamination and work activities. Monitor at work areas and, where necessary, site boundaries. Stop or modify activities if monitoring indicates unacceptable releases.
- Maintain spill kits suitable for acid and contaminated liquids. Establish procedures for acid spills, Pb-containing dust releases, contaminated-water releases and damaged batteries. Ensure emergency response arrangements remain operational throughout decommissioning.
- Package and label hazardous materials appropriately and use licensed/ authorised transporters where required. Cover loads and prevent leakage or dust emissions during transport. Maintain records showing destination and final treatment/recycling/ disposal.
- Sequence demolition so that hazardous materials and contaminated equipment are removed before general demolition. Isolate contaminated areas, suppress dust and prevent contaminated demolition rubble from being mixed with “cleaner” waste.

- Treat disposable PPE, filters, wipes, absorbents and cleaning materials from contaminated areas as potentially Pb-contaminated until characterised. Prevent workers from taking contaminated clothing or equipment off-site. WHO specifically highlights the risk of workers transferring Pb contamination from workplaces to homes.
- After dismantling, conduct systematic cleaning using wet methods and/or HEPA vacuuming. Remove residual Pb-bearing deposits from floors, drains, structural surfaces and equipment. Undertake confirmatory environmental sampling to demonstrate that applicable cleanup criteria have been achieved.
- Monitor relevant parameters throughout decommissioning, potentially including airborne particulate/ Pb, soil Pb, surface water, groundwater, wastewater pH and metals, and waste characteristics. Monitoring should be risk-based and linked to the identified contamination pathways.
- Prepare a final environmental closure report documenting decommissioning activities, quantities and destinations of hazardous materials/ wastes, analytical results, remediation undertaken, incidents, monitoring data and confirmation that environmental closure requirements have been met. The International Finance Corporation's General Environmental, Health, and Safety Guidelines specifically identify contaminated land, hazardous-material management, waste management, wastewater, air emissions and decommissioning as relevant environmental-management areas.
- Develop a formal workforce-transition plan well in advance of closure. Provide affected employees with written notice, information on severance benefits, pension/ provident-fund arrangements and their legal rights.
- Provide affected workers with financial counselling, budgeting assistance, and debt-management support. Where feasible, phase workforce reductions to give employees time to seek alternative employment.
- Implement retraining and skills-development programmes targeted at affected workers, including training in electrical work, mechanical maintenance, waste management, environmental monitoring, logistics, construction and other locally marketable skills.
- Prioritise local recruitment for decommissioning, demolition, transportation, rehabilitation, security, landscaping and environmental-monitoring activities, provided workers meet competency requirements.
- Identify local suppliers and contractors dependent on the facility and provide advance information about closure. Where possible, use local businesses for decommissioning-related procurement.
- Develop a local economic-transition strategy, potentially including support for small businesses, enterprise development, skills programmes and linkages with other employers in the area.
- Engage the relevant municipality regarding anticipated changes in the site's economic contribution and investigate opportunities for alternative productive reuse of the site.
- Establish a structured stakeholder engagement programme. Clearly communicate the closure schedule, reasons for closure, employment implications, environmental risks, rehabilitation activities and future land-use plans.
- Establish a transparent grievance mechanism through which employees, contractors, and community members can submit complaints and receive documented responses.
- Identify social investments previously supported by the facility and develop a transition plan with beneficiaries, rather than abruptly terminating critical programmes.
- Provide specialised training, PPE, medical surveillance, safe-work procedures and competent supervision for workers potentially exposed to Pb, H₂SO₄, contaminated dust, residues, and other hazardous materials.
- Secure the site, control public access, and ensure that Pb-containing materials, spent batteries, contaminated soil, and hazardous wastes are properly contained, transported, and treated/ recycled by authorised facilities.
- Identify informal traders, service providers, and other vulnerable groups that may have benefited from the facility and consider appropriate transition measures.
- Develop a traffic-management plan for trucks transporting equipment, scrap metal, batteries, and hazardous waste. Schedule heavy vehicle movements to minimise disruption to surrounding communities.
- Ensure prompt site clean-up, remediation, rehabilitation, and appropriate redevelopment/ reuse so that the closed facility does not become a source of blight or perceived contamination.
- Conduct contamination assessments and remediate areas affected by Pb, acid, hydrocarbons or other hazardous substances. Maintain appropriate post-closure monitoring where contamination remains.
- Retain suitably qualified personnel during critical decommissioning stages to ensure safe shutdown, hazardous-

material identification, equipment isolation, and knowledge transfer.

- Communicate closure decisions honestly and consistently, including the expected duration, employment measures, environmental safeguards, and proposed future use of the site.
- Conduct a vulnerability assessment and give particular attention to low-income households, single-income households, and workers with limited transferable skills.
- Ensure that adequate financial provision is available for decommissioning, rehabilitation, and post-closure obligations. South African environmental legislation provides for financial provision for specified rehabilitation, decommissioning, closure and post-closure activities.

2.4.4. No-Go

- Continue to implement the approved EMPr.

3. CLIMATE CHANGE ASSESSMENT

Climate change issues must be considered as part of the EIA process. EAP must determine:

- a) the potential impact of climate change on society and the economy, whether the impact is negative or positive, considering that society needs to be at the centre of the proposed development;
- b) The potential alternatives of the proposed development, alternatives that will have less impact on climate change (environment and generation of waste included), the society and economy;
- c) whether, and to what extent, the proposed development will result in the release of GHG emissions;
- d) whether the proposed development is necessary to achieve long term decarbonisation goals;
- e) the impact of the development on social, economic, natural, and built environment that are crucial for climate change, adaptation and resilience;
- f) the projected impact of climate change on proposed development and surrounding environment, and implications for the development;
- g) explanation of how the impacts are likely to be exacerbated or minimised as result of climate change and what measures are likely to be implemented to accommodate and manage (adapt to) the anticipated worst scenario where applicable; and
- h) whether, and to what extent, the impacts identified in (a) to (g) can be mitigated.

3.1. General Potential Impact of Climate on Society and the Economy

Climate change has the potential to negatively affect society and the economy through increased temperatures, extreme weather events, water scarcity, and disruptions to infrastructure and supply chains. These impacts may affect employee health and safety, increase operational costs, reduce productivity, and place additional pressure on surrounding communities. The economic and societal impacts associated with climate change, not necessarily the proposed BM Line 3 addition, are discussed below.

Economic Impacts:

- Extreme weather events: Climate change has led to an increase in the frequency and intensity of extreme weather events. The costs associated with rebuilding and recovery divert resources from essential sectors like education and healthcare.
- Supply chain disruptions: Changes in temperature and precipitation patterns can severely affect agricultural yields, leading to food shortages and price increases. Droughts can reduce water availability for industrial production, impacting manufacturing and other water-intensive sectors. This scarcity can lead to higher production costs and inflation, ultimately reducing economic competitiveness.
- Long-term GDP decline: Research indicates that each 1°C rise in global temperature could reduce world GDP by 1 to 3%. More recent studies suggest that the economic damages of climate change may be significantly underestimated, with potential declines in GDP of 30 to 50% by the year 2100 if temperatures rise by 2°C (Havard Magazine, 2024).

Societal Impacts:

- Health risks: Climate change exacerbates health risks through increased heatwaves, air pollution, and the spread of vector-borne diseases. Vulnerable populations, including the elderly and those with pre-existing health conditions, are particularly at risk.
- Food security: Disruptions in agricultural productivity due to changing climate conditions threaten food security. Crop failures and reduced yields can lead to higher food prices and increased hunger, particularly in developing regions.
- Infrastructure strain: Rising sea levels and extreme weather events threaten coastal infrastructure, necessitating costly adaptations or relocations. This can strain public resources and impact economic stability.
- Social displacement: Climate change can lead to the displacement of communities due to extreme weather events or uninhabitable conditions, creating climate refugees and increasing social tensions.

The consequences of climate change are profound and multifaceted, affecting economic stability, public health, food security, and social cohesion. Addressing these challenges requires urgent action to mitigate climate change and adapt to its inevitable impacts, ensuring a resilient future for society and the economy. The impact assessment above addresses potential climate change impacts associated with BM Line 3.

The assembly of lead-acid batteries also supports energy storage applications that improve electricity reliability and backup power, contributing positively to community resilience and economic stability. Ensuring that local employment, worker welfare, and environmental protection remain central to the development is essential to achieving sustainable outcomes.

3.2. Potential Alternatives with lower Climate, Environmental, Social, and Economic Impacts

The assembly and use of lead-acid batteries contribute to climate change through the release of Pb and H₂SO₄, which are harmful to the environment and human health. Potential alternatives include improving energy efficiency within the assembly process, using renewable electricity to power the facility, sourcing recycled Pb instead of primary Pb where feasible, and implementing waste minimisation and recycling programmes. Alternative battery technologies, such as lithium-ion batteries, may offer lower life-cycle emissions in certain applications but present their own environmental and resource challenges. Optimising the existing lead-acid battery assembly process through cleaner production practices and responsible waste management can significantly reduce environmental impacts while maintaining social and economic benefits such as employment and local manufacturing.

3.3. Extent to which the proposed BM Line 3 will result in GHG Emissions

The proposed lead-acid battery assembly facility will generate GHG emissions, albeit indirectly, primarily through electricity consumption, transportation of raw materials and finished products, and the indirect emissions associated with the production of battery components. These are all components of the larger process, which is already authorised, not the BM Line 3 itself. As the facility focuses on battery assembly rather than primary Pb smelting, direct process emissions are expected to be relatively low. Overall emissions can be further reduced through energy-efficient equipment, renewable energy integration, and improved logistics planning.

3.4. Necessity of the proposed BM Line 3 to achieve Long-Term Decarbonisation Goals

The proposed BM Line 3 addition supports long-term decarbonisation by supplying batteries that could enable energy storage for renewable energy systems, backup power applications, traditional and electric vehicle transportation, telecommunications, grid stability, and any other applications battery purchasers utilise batteries for. Lead-acid batteries remain an important and highly recyclable energy storage technology in many sectors. While they are not a zero-emission technology themselves, their role in supporting renewable energy integration and improving energy resilience contributes positively to broader climate change mitigation objectives.

3.5. Impact of the development on the social, economic, natural, and built environment

The proposed development is expected to have positive social and economic impacts through job creation, skills development, and support for local industries. Potential impacts on the natural environment include energy use, waste generation, and the handling of hazardous materials such as Pb and battery electrolyte, which require strict environmental

management. Impacts on the built environment are expected to be limited to normal industrial activities and increased transport movements. With appropriate pollution prevention, waste management, and environmental monitoring measures, these impacts can be effectively managed while strengthening local economic resilience.

3.6. Projected impact of climate change on the proposed battery assembly line addition and surrounding environment

Climate change may affect the proposed BM Line 3 through increased temperatures, heavier rainfall events, flooding, water shortages, and disruptions to electricity supply and transportation networks. These factors could interrupt operations, damage infrastructure, or affect employee safety. The surrounding environment may also experience increased vulnerability to erosion, stormwater runoff, and extreme weather events, requiring appropriate site design, drainage systems, and emergency preparedness measures to ensure continued operational resilience.

3.7. How impacts may be exacerbated or minimised by climate change and adaptation measures

Climate change could exacerbate operational risks by increasing heat stress on workers, raising cooling requirements, intensifying stormwater management challenges, and increasing the likelihood of supply chain disruptions. These risks can be minimised through improved building design, improved drainage infrastructure, energy-efficient equipment, backup power systems, water conservation measures, heat stress management programmes, emergency response planning, and the incorporation of renewable energy where practical. Regular climate risk assessments and adaptive management will further strengthen the facility's resilience under future climate scenarios.

3.8. Extent to which identified impacts can be mitigated

Most of the identified climate-related and environmental impacts can be mitigated to acceptable levels through the implementation of appropriate design, operational, and environmental management measures as discussed above. These include adopting energy-efficient technologies, increasing the use of renewable electricity, minimising waste generation, maximising battery recycling and material recovery, continuous implementation of effective hazardous waste management procedures, reducing transport emissions where possible, and maintaining comprehensive environmental monitoring and emergency response plans. Although residual impacts associated with energy use and transportation cannot be completely eliminated, they are expected to remain low to moderate and manageable throughout the operational phase.

To mitigate these impacts, it is essential to continuously implement stringent regulations, effective recycling initiatives, and responsible disposal methods. Auto-X is committed to continue using sustainable manufacturing processes and promoting the recycling of lead-acid batteries to reduce their environmental footprint.

4. ENVIRONMENTAL IMPACT STATEMENT

Taking the assessment of potential impacts into account, please provide an environmental impact statement that summarises the impact that the proposed activity and its alternatives may have on the environment after the management and mitigation of impacts have been taken into account, with specific reference to types of impact, duration of impacts, likelihood of potential impacts actually occurring and the significance of impacts.

Alternative A (preferred alternative)

A conservative approach was followed by the air quality specialist in estimating emissions from the operation of the battery manufacture and assembly facility. The rationale behind this approach was that the outcomes will be an overestimation of the impact on air quality. Should this impact prove negligible the real impact would be even less. Three emission scenarios were assessed against that approach: the licensed emissions on its own, the actual expected plant emissions, and a cumulative assessment based on the actual emissions of the plant and surrounding sources.

On the measured and expected basis, the model shows that the annual average Pb concentration will exceed the NAAQS, but that the exceedance is confined to the immediate vicinity of the plant. Periodic ambient Pb monitoring must be considered in the easterly corner of the facility or neighbouring property to verify Pb emissions against the NAAQS.

The summary of the potential impacts identified, after the application of management and mitigation measures, along with their type, duration, probability (likelihood of occurrence), and significance are provided in **Table 5**.

The findings of the specialist study undertaken together with the broader BA process conclude that there are no fatal flaws that should prevent the BM Line 3 installation and operation from proceeding. However, the key potential impacts (**Table 5**) have been identified which will require the application of activity specific mitigation measures. These mitigation measures are included in this report and within the Draft EMP to ensure that they receive the necessary attention.

Based on this section of the report, the proposed BM Line 3 will have some contribution to local economic upliftment. The potential negative impacts identified were not found to be highly significant and with appropriate mitigation can be reduced to insignificant or very low though negative. The potential positive impacts ranged from medium to high in terms of significance, provided that the enhancement measures provided are applied.

Table 5: Summary of potential impacts, post mitigation

Impact	Duration	Consequence	Probability	Significance
Planning and Design				
Job opportunities	Short-term	Medium	Definite	Medium
BM Line 3 permitting requirements	Temporary	Very low	Improbable	Insignificant
Source energy efficient machinery	Long-term	Medium	Definite	Medium
Operation				
Release of Pb and H ₂ SO ₄ into the environment	Long-term	Very low	Probable	Very low
Release of fumes/ air emissions due the welding/ burning associated with BM Line 3	Long-term	Very low	Probable	Very low
Energy consumption	Long-term	Very low	Probable	Very low
GHG emissions	Long-term	Very low	Probable	Very low
Job retention	Long-term	Medium	Definite	Medium
Environmental pollution, including air pollution, and health risks associated with Pb, H ₂ SO ₄ , and fumes	Long-term	Very low	Possible	Insignificant
Localised, around some machinery, elevated noise levels	Medium-term	Very low	Possible	Insignificant
Skills development and transfer	Long-term	Medium	Definite	Medium
Employment and skills development associated with battery recycling and other second-life applications	Long-term	Medium	Definite	Medium
Release of Pb and H ₂ SO ₄ into the environment (cumulative)	Long-term	Very low	Possible	Insignificant
Production of H ₂ SO ₄ can cause acid rain which has a negative effect to soil, plants, and aquatic life	Medium-term	Very low	Possible	Insignificant
The mining of Pb can result in environmental degradation	Long-term	Very low	Possible	Insignificant
Depletion of finite resources, i.e. Pb alloy	Long-term	Very low	Possible	Insignificant
Decommissioning and Closure Phase				
Free up of energy for other uses	Medium-term	Medium	Probable	Medium
Job losses	Medium-term	Very low	Probable	Very low
Negative effect on the lead-acid battery market	Medium-term	Very low	Probable	Very low
Lessened environmental pollution, including air pollution	Medium-term	High	Probable	High
Skills and skills transfer loss	Medium-term	Very low	Possible	Insignificant
Job losses (cumulative)	Medium-term	Very low	Probable	Very low
No-Go				
Status-quo; no BM Line 3, but continuation of BM Lines 1 and 2	n/a	n/a	n/a	n/a
No additional job creation, skills development and transfer, and economic inflow	n/a	n/a	n/a	n/a
The other associated impacts will still continue as per the existing operations, save for the economic impacts	n/a	n/a	n/a	n/a

No-go alternative (compulsory)

As machinery ages its efficiency becomes reduced. This is the case with battery assembly machinery as well which is employed by the applicant in their day-to-day activities. The reduced efficiency has a few implications, i.e. higher energy consumption than newer/ latest technology machinery; frequent requirement for maintenance which means increase in down-time and decrease in production; etc. To circumvent these issues, Auto-X proposes the addition of BM Line 3 which is essentially a replacement of an aged line within the facility.

The no-go alternative assumes that the operations continue as are, no new machinery installation and operation. This means the acceptance of higher energy consumption (in comparison to use of newer machinery) and higher facility down-time which does not only have implications for the applicant but for the economic standing of the employees and the local battery market. Notwithstanding the above, the potential negative impacts that may be associated with BM Line 3 operation will not occur should the CA choose the no-go decision.

As indicated above, the assessment that has been conducted by both the specialist and the *Reg. EAP* indicate that there is no potential fatal flaw in the operation of BM Line 3. Thus, the *Reg. EAP* does not recommend the no-go alternative.

SECTION E: RECOMMENDATIONS OF PRACTITIONER

Is the information contained in this report and the documentation attached hereto sufficient to make a decision in respect of the activity applied for (in the view of the EAP)?

YES	
YES	

Is an EMPr attached?

The EMPr must be attached as **Appendix F**.

~~If "NO", indicate the aspects that should be assessed further as part of a S&EIA process before a decision can be made (list the aspects that require further assessment):~~

If "YES", please list any recommended conditions, including mitigation measures that should be considered for inclusion in any authorisation that may be granted by the CA in respect of the application:

In summary and based on this detailed assessment and that of the specialist, it is the EAP's opinion that the BM Line 3 installation and operation can be authorised, with conditions and routine inspections stipulated in the EA. This will ensure that all potential negative impacts are mitigated and monitored efficiently.

It is the conclusion therefore that, with the implementation of the recommended mitigation measures included in this report and in the Draft EMPr, all of the identified potential impacts can be mitigated to an acceptable level. Of these measures, the most important are:

- On the licenced (AEL) basis the predicted maximum exceeds its criterion for every pollutant and averaging period assessed. The issue remains that the AEL and impact assessment must consider the fact that an operator is allowed to emit up to the maximum permitted concentration and the applied flow rate (although this is not the case as per the measured emissions, but allowance of the issued AEL). It is therefore critical that the information pertaining to the AEL is accurate and relevant in terms of the process classifications, controlled pollutants, and flow rates. To reduce the predicted air quality impact the emissions allowed for in the AEL must either be reduced or the release height of the emissions must be increased to allow for better dispersion before the pollutants reach ground-level. To reduce the allowed emissions, it is recommended that the listed stacks, and their associated processes, are carefully evaluated to determine the applicability of the Listed Activity to those processes. Additionally, the maximum allowed emissions may be reduced by the Licensing Authority, who is duly authorised to do so by the NEMAQA, via the introduction of stricter emission limits than the nationally listed emission limits stated in the List of Activities and associated Minimum Emission Standards. Alternatively, release heights can be improved by extending stack heights, thus allowing better dispersion of pollutants. A repeated dispersion modelling study will be required to obtain the minimum stack heights required to prevent exceedance of the NAAQS levels.

LAQS (2024) recommends that a revised AEL be issued to Auto-X, however, periodic ambient Pb monitoring must be considered in the easterly corner of the facility or neighbouring property to verify Pb emissions against the NAAQS. Auto-X acknowledges the difference between licenced maximums and actual measured emissions. The AEL variation application submitted to NMBM includes corrections to reflect actual measured stack parameters, ensuring the licence accurately reflects plant operations.

- The currently practiced stack emissions sampling should continue as the facility remains operational.

Planning and Design Phase:

- Ensure any new job opportunities accrue to the local community, where feasible.

Operational Phase:

- Minimise the emissions of Pb, H₂SO₄, and any other materials utilised within the facility.
- Regular testing/ sampling of Pb levels to ascertain emissions are minimised. Periodic ambient Pb monitoring must be considered in the easterly corner of the facility or neighbouring property to verify Pb emissions against the NAAQS.
- Strengthen policies and plans to prevent Pb exposure, including the adoption of standards and strong implementation of these measures.
- Optimise industrial processes, where feasible, and implement smart technologies for demand management.
- Conduct comprehensive energy audits to identify high-consumption areas and implement targeted efficiency upgrades.
- Encourage management and workers behaviour changes and appliance upgrades to minimise demand.
- Continue to conduct comprehensive lifecycle assessments to quantify and mitigate GHG footprints associated with battery production, including material sourcing and manufacturing.
- Continue with the extended producer responsibility practices currently implemented.
- Promote responsible disposal practices to minimise environmental harm.
- Continue to adhere to policies and regulations governing the proper disposal and recycling of lead-acid batteries.
- Implement risk mitigation strategies, such as proper storage, transportation, and treatment of spent batteries, along with public education and awareness campaigns. This helps minimise environmental risks and ensure responsible battery management.
- Adhere to the EMPr and ensure that contractors are trained in environmentally responsible practices.
- Continue with the monitoring of air quality, correct PPE use, harvesting rainwater, implementing energy saving measurements, implementing water saving measures, implementing air pollution abatement measures, etc.
- Continue to continuously engage with employees on the safety and health issues, i.e. the current use imagery around the facility and the toolbox talks.
- Continue with the monitoring of GHG emissions.
- Continue with the developed implementation of scheduled preventative maintenance programme.
- Conduct regular inspections using emissions leak detection technology.

SECTION F: APPENDICES

The following appendixes must be attached as appropriate:

Appendix A: Site plan(s)

Appendix B: Photographs

Appendix C: Facility illustration(s)

Appendix D: Air Quality Report

Appendix E: CRR

Appendix E1: Register of I&APs

Appendix F: EMPr

Appendix G: Other information

Appendix G1: Locality Map and Other Maps

Appendix G2: Copy of AEL Variation Application

Appendix G3: Proof of submission of AEL review application

Appendix G4: SSVr

Appendix G5: EAP Application, CV, and EAPASA Certificate

Appendix G6: Landowner Consent Letter

Appendix G7: DEDEAT Correspondence

Appendix A: Site plan(s)

Appendix B: Photographs

Appendix C: Facility illustration(s)

Appendix D: Specialist report/ Air Quality Impact Assessment Report

Appendix E: CRR (This can only be made available post PPP as its development is based on comments still to be received)

Appendix E1: Register of I&APs

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